

Graded Containment with an Observability Price: Sequential Response to Ambiguous Trading Incidents

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Abstract

When an anomaly appears on an electronic trading gateway, the operator often cannot tell whether it comes from a compromise or from a fault, and the usual response is a kill-switch. We model the response as a finite-horizon partially observed Markov decision process over the hidden cause, with a graded ladder of actions (monitor, throttle, verify on a separate channel, kill, leave), solved exactly on a belief grid, with a benign market shock as an optional third cause. Throttling reduces harm but also blinds: it multiplies the per-period order-flow separation by a discriminability factor $\kappa \in [0, 1]$. Classical POMDP structure, specialised to the model, gives a single critical factor κ^* below which an optimal operator need never throttle, and recovers the one-shot verification rule of our companion paper when order flow is uninformative. Two results are specific to graded containment: $\kappa^* > 0$ whenever throttling has a disruption cost; and, in a diffusion version of the model with a disruption cost or residual harm, the throttle is worth using at a belief exactly when its running cost per unit of information is below monitoring's, which gives κ^* in closed form. An exact computation, with criteria fixed before any result and calibrated on the cells of the companion identifiability study in which the classifier has no supported improvement over the best constant policy, supports three claims: the graded policy beats the best Bayesian kill-switch materially at a share 0.796 of the converged points, with little of the gain from the throttle; $\kappa^* < 1$ on a small part of the region; and there an operator who ignores the observability price does worse than the kill-switch. The gain over the one-shot rule is material at only a share 0.183

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of the points, which weakens that claim; the effect of a benign third cause is inconclusive at this power; and the claim that the diffusion law locates the discrete κ^* is refuted; separately, a closed-form policy derived from the law recovers most of the graded gain.

Keywords: OR in security, partially observed Markov decision process, sequential decision making, graded containment, value of information, incident response, electronic trading

1. Introduction

An alarm on an electronic trading gateway rarely comes with its cause attached. The same burst of abnormal orders can be the work of an adversary acting through a legitimate session (a *compromise*) or the symptom of a benign malfunction (a *fault*). The two call for opposite responses: a compromise should be contained quickly, while containing a fault disrupts a legitimate client for nothing. The operator’s standard tool is a kill-switch, which stops the flow once the evidence crosses a threshold. Between doing nothing and killing, operators also have intermediate options: they can throttle the flow, which limits harm while the incident continues, or check the activity on a channel that order flow does not carry, for instance by validating the session’s credentials or calling the client.

This paper asks when such a graded response is worth more than the best kill-switch, and what it costs. The answer turns on a feature that a static decision rule cannot see.

Containment changes what the operator observes next. Paper 1 (Mejrissi, 2026b) gives the starting point. When the incident’s injection enters the latent state of the book, Paper 1’s Theorem 2 shows that two mechanism instances are observationally indistinguishable at every time after the onset if and only if their injection difference $\Delta_0 = \theta_K b_K - \theta_F b_F$ lies in the unobservable subspace of (A, C) . Its Theorem 3, the finite-horizon version of Theorem 2, gives the same statement over H post-onset observations, with the condition $CA^j \Delta_0 = 0$ for $j = 0, \dots, H - 1$. A pair that gives the same immediate output ($C\Delta_0 = 0$, condition $(*)$ of Paper 1’s Theorem 1) can therefore still be told apart through the feedback of the injection into the state. Theorem 2 is an equivalence about equality of laws. It contains no containment action and no measure of how much discriminability is kept. Both come from this paper. The measure is the *discriminability factor* κ :

the share of the per-period order-flow separation between the causes that a throttled period keeps (Definition 3). We model two throttles in Paper 1’s linear-Gaussian model (Definition 11): one that scales the injection by a factor r , and one that suppresses feedback, keeping the injection in the observation and removing it from the state recursion, so that throttled periods follow Paper 1’s output-level model. Proposition 4 then shows four things, one for each of its parts:

- (i) a throttle that scales the injection by r from the onset gives $\kappa = r$, so the throttle’s harm reduction and its loss of discriminability are tied.
- (ii) for a pair confounded at the output level, a feedback-suppressing throttle in force from the onset gives both causes the same law of observations. All the pair’s discriminability came from feedback, so $\kappa = 0$.
- (iii) if the throttle starts after the onset, (i) and (ii) can fail: the feedback accumulated before the throttle starts leaves a residual term, which decays according to the dynamics.
- (iv) for pairs that are not confounded at the output level, the linear model does not determine whether κ is below or above one: there are explicit instances with $\kappa < 1$, $\kappa = 1$ and $\kappa > 1$.

That throttling costs discriminability ($\kappa \leq 1$, Assumption 4) is therefore derived from Paper 1’s model only for the scaling throttle and, for pairs confounded at the output level, for the feedback-suppressing throttle in force from the onset. Otherwise it is an assumption of this paper’s model, not a prediction of Paper 1 (Proposition 4(iv)).

We study the response as a finite-horizon partially observed Markov decision process (POMDP) over the hidden cause (Section 3). The operator holds a belief that the cause is a compromise and, at each period, chooses one of five actions: **monitor**, **throttle**, **verify** (the costed, delayed check of Paper 2 (Mejrissi, 2026c)), **kill** or **leave**. The model is solved exactly by backward induction on a belief grid, without reinforcement learning or function approximation. In the baseline the cause is a compromise or a fault; in the extension a benign market shock is a third candidate cause, and the belief lives on the 2-simplex.

Contributions.

- *Model.* A POMDP in which containment has two effects at once: it scales the harm rate by a harm factor ρ and the per-period order-flow separation by the discriminability factor κ (Definitions 1–12, Assumptions 1–5). Both enter the POMDP in Section 3, κ in Definition 3 and ρ in Definition 6.
- *Where $\kappa < 1$ comes from* (Proposition 4, Section 3.7): the two throttles of Definition 11 in Paper 1’s linear-Gaussian model, with explicit instances for each case.
- *Structure* (Section 4): classical results of POMDP theory, specialised to this model and not claimed as new: concavity of the value function (Lemma 1, Proposition 1); monotonicity of the optimal value in κ and in d , which makes the critical discriminability factor κ^* a single threshold below which an optimal operator need never throttle (Theorem 1); kill and leave regions that are intervals (Proposition 2); and the exact embedding of Paper 2’s one-shot rule when order flow is uninformative (Proposition 3).
- *A blinding throttle is never used* (Proposition 6, Section 4.5): a throttle that removes all evidence is never strictly useful, and with a disruption cost $\kappa^* > 0$, in the two-state and the three-state model.
- *A cost-per-information law* (Proposition 5, Section 4.6): in a diffusion version of the two-state model, the throttle is worth using at a belief exactly when its running cost per unit of information is below monitoring’s. This gives κ^* in closed form; it shows that a throttle that scales the flow is worth using only if harm falls faster than quadratically in the flow; and it shows that the cost of an operator who plans as if throttling kept full discriminability grows without bound as κ falls. The technique, a time change to the information clock, comes from sequential testing with a controlled signal (Campbell et al., 2025); the belief-dependent switch, the action that acts on harm and on information at once, the closed form of κ^* and the result on the blind operator are new. The law holds when the throttle has a disruption cost or leaves some harm; without either, a free and informative throttle leaves no optimal policy.
- *A numerical test, fixed in advance* (Section 5), on a region calibrated from the set $\mathcal{F}$ of Paper 1’s cells without a supported improvement over

the best constant policy, with the design, the tolerance, the materiality margin and the criteria that would weaken or refute each claim fixed before any result. Its outcomes, positive and negative: the graded policy has strictly and materially lower expected cost than the best Bayesian kill-switch at most points (supported), with little of the gain from the throttle; $\kappa^* < 1$ on a non-empty but small part of the region, non-decreasing in the disruption cost and the harm factor (supported); there, an operator who plans as if throttling kept full discriminability does worse than the kill-switch (supported); the gain over Paper 2’s one-shot rule, a corollary of Proposition 3, is material at no more than half of the points (weakened); the effect of a benign third cause is inconclusive at this power. Under a second brief, the claim that the law of Proposition 5 locates the κ^* of the discrete, finite-horizon model is refuted; its predictions for scaling throttles and for the band order of the optimal policy hold, its prediction $\kappa^* = \sqrt{\rho}$ without disruption cost is weakened, and a closed-form policy derived from it recovers most of the graded gain.

The comparison is against the *best* Bayesian kill-switch, an optimal stopping rule on the belief, not a fixed threshold. Weak dominance of the graded policy holds by construction, since the kill-switch is in its policy class; it is not claimed as a finding. What the paper claims is *strict* dominance on a stated region, with the gain split into what verification and what throttling contribute. Section 2 positions the paper; Section 6 discusses scope and limits.

2. Related work

POMDP intrusion response. The closest precedent is the treatment of intrusion prevention as an optimal stopping problem by Hammar and Stadler (2022): a defender with several progressively stronger stops faces a partially observed intrusion, and the optimal policies are thresholds in the belief. Our setting differs in three ways. First, the hypotheses there are intrusion against no intrusion, whereas ours are compromise against fault (and, in the extension, a benign shock): both hypotheses are anomalies, and the question is their cause. Second, there is no verification on a separate channel. Third, a defensive action there does not degrade what the defender observes next, while in our model throttling multiplies the order-flow separation by κ . The

same authors extend the stopping formulation to a dynamic attacker and learn near-optimal responses by self-play (Hammar and Stadler, 2024); our cause is non-strategic (Assumption 5). Miehl, Rasouli and Teneketzis (Miehl et al., 2018) model the defence of a network against an attacker progressing on an attack graph as a POMDP, in which the defender trades security against availability. There the belief is over the attacker’s progress; here it is over the cause of an anomaly, a scalar or a point of the 2-simplex, and the operator’s actions change the quality of observation. Kazeminajafabadi and Imani choose defence actions and the allocation of limited monitoring jointly in a POMDP over Bayesian attack graphs (Kazeminajafabadi and Imani, 2024), and derive optimal monitoring and attack-detection policies when only part of the network can be monitored (Kazeminajafabadi and Imani, 2023); monitoring there is a separate, budgeted resource, whereas in our model observability falls as a side effect of the containment action. On-line conjecture revision lets a defender adapt its response when its model of the system and of the attacker is misspecified (Hammar et al., 2025); we assume a known model and solve it exactly, and study the structure of the cost/observability trade-off rather than learning. Exactness is not itself the contribution: a causal POMDP model of the CAGE-2 benchmark already admits an online tree-search method with a provably optimal defender strategy (Hammar et al., 2026). Probing a live IT system to identify its causal model has also been treated as an online decision problem in which the intervention policy is meant to interfere little with operations (Hammar and Stadler, 2026); there the price is paid for probing, here it is paid for mitigating. Surveys of intrusion response for cyber-physical systems (Bashendy et al., 2023) and of deep reinforcement learning for cyber security (Nguyen and Reddi, 2023) cover the decision methods (MDP and POMDP models, games, learned policies) used to automate response. Partial containment has been posed as an optimal control problem: for malware in IoT networks, an intermediate immediate-response state before full patching is balanced against patching cost (Jafar et al., 2024); propagation there is fully observed, the cause is not in question, and the partial action carries no information cost.

Verification and one-shot diagnosis. The verify action is the costed, delayed check of Paper 2 (Mejrissi, 2026c), which chooses once between leaving, verifying and containing. Our model makes that choice sequential and adds order flow and throttling; Proposition 3 shows that Paper 2’s rule is recovered exactly when order flow is uninformative and the horizon is at least the

verification delay ($d = 0$, $T \geq \ell_v$).

Active hypothesis testing and dual control. Choosing observations to learn a hypothesis is the subject of the sequential design of experiments (Chernoff, 1959) and of active and controlled-sensing hypothesis testing (Naghshvar and Javidi, 2013; Nitinawarat et al., 2013). For controlled sensing with more than two hypotheses, the error exponent is governed by the best Chernoff information over controls (Nitinawarat et al., 2013); the discriminability factor κ enters exactly such a quantity. With a fixed number of experiments, a test may end by declaring a hypothesis or “inconclusive” (Kartik et al., 2022), the closest counterpart of our finite horizon with a non-committal terminal **leave**. Recent work treats sequential anomaly detection when only some sources can be sampled at a time (Tsopelakos and Fellouris, 2023), quickest change detection with a constraint on the cost of observations, where an on-off observation control is asymptotically robust optimal (Hou et al., 2025), and quickest change detection with experiments of different cost (Lubenian and Banerjee, 2026). There, actions buy information, and skipping an observation saves its cost without reducing harm. In our model the most informative posture, **monitor**, is also the most harmful under a compromise, and the action that reduces harm, **throttle**, reduces information: the throttle has a dual role through ρ and κ together. That the value of an experiment is monotone in its informativeness is Blackwell’s ordering (Blackwell, 1953); structural results of this kind for POMDPs and controlled sensing are collected in (Krishnamurthy, 2025). Theorem 1 is an instance of this technique applied to the throttle’s discriminability factor, and we do not claim the technique as new. What the paper adds is where the resulting threshold κ^* lies: strictly above zero whenever throttling has a disruption cost (Proposition 6), in closed form in a diffusion model (Proposition 5), and on a calibrated region by computation; and what an operator loses by ignoring it.

Cost of information and intensity of experimentation. Proposition 5 rests on a technique from sequential testing with a controlled signal. For a Brownian drift whose signal intensity the observer controls at a running cost that does not depend on the belief, Campbell et al. (2025) show that the optimal control is the constant minimiser of the cost per unit of information rate, which reduces the problem to optimal stopping; we use their time change to the information clock. In operations research, Harrison and Sunar (2015) study a binary unknown that can be learnt through several modes, each with a

cost rate and a signal quality, before a decision to invest or abandon, and in economics Moscarini and Smith (2001) study the choice of how intensely to experiment before a decision. The active hypothesis-testing literature ranks experiments by the information they yield per sample (Chernoff, 1959; Naghshvar and Javidi, 2013); for two Gaussian hypotheses whose means differ by κd , that information is proportional to $\kappa^2 d^2$, so κ^2 is the relative information rate of a throttled period. In Campbell et al. (2025), in Harrison and Sunar (2015) and in active hypothesis testing, the cost of an experiment does not depend on the belief and reduces no harm. In our model the harm accrues only under a compromise, so the throttle’s cost per unit of information depends on the belief, and the throttle is an experiment that also contains: the optimal posture switches with the belief, and the switch point gives κ^* . In dual control, inputs both steer and probe, and cautious control can switch learning off (Mesbah, 2018); the observability price of claims 3 and 4 (Section 5.1) is a discrete incident-response analogue of this turn-off effect, in which containment removes evidence and the problem is solved exactly on a small belief space.

Active fault diagnosis and the price of observability. Active fault diagnosis designs inputs to tell faults apart under a decision cost. Straka and Punčochář (2024) generate excitation inputs in a multiple-model framework with costs that penalize missed detections, false alarms and wrong identifications jointly; related work designs inputs that reconcile detection with the control objective (Cao et al., 2024) and integrates diagnosis with fault-tolerant control under input constraints (Guo and He, 2026). Our cause set contains an attack, and our actions reduce harm as well as shaping what is observed. Telling attacks from faults is itself a detector-design problem: switching sensor watermarks can detect threats and discriminate physical faults from man-in-the-middle attacks (Zhang et al., 2026), with no cost of response; we decide how to respond while that discrimination is unresolved. The mirror image of our throttle is physical watermarking, in which control performance is paid to make attacks detectable and the trade-off is optimized (Mo et al., 2015); a watermark can be designed to maximize the Kullback–Leibler divergence for a given increase in LQG control cost (Naha et al., 2023), an explicit price of observability. Moving-target perturbations raise the same question of when a defensive action makes attacks detectable (Liu et al., 2022). Our throttle runs the other way: it buys harm reduction and pays in discriminability.

Kill-switches and circuit breakers. Kill-switch and circuit-breaker practice on trading venues stops activity on a fixed trigger. Hard triggers have side effects: market-wide price-triggered halts create a magnet effect and have an ambiguous welfare effect (Chen et al., 2024), a price model with halts reproduces the magnet effect (Lera et al., 2024), and intraday evidence around China’s 2016 market-wide halts shows herding and volatility causing each other (Wang et al., 2022). A forward-looking, belief-dependent breaker can outperform a static one (Bongaerts et al., 2024); that result concerns market-level design against runs, not the diagnosis of a cause at one firm’s gateway, and we read it as a parallel. Our kill-switch baselines (Definition 9) are the fixed-threshold rule KS_θ and the best Bayesian stopping rule KS^* ; the latter is the strongest version of the kill-switch that the model allows, and it is the one the graded policy is compared with.

3. Model

An operator watches an anomaly on a trading gateway. The anomaly has one of two causes, a compromise K or a fault F ; in the extension, a third cause B , a benign market shock, is also possible. The operator does not see the cause. At each period the operator observes an order-flow statistic and chooses an action, until a terminal action ends the episode or the horizon is reached. This section specifies the decision problem completely: time and states, actions, observations, verification, belief update, costs, the Bellman recursion and its numerical solution, the baseline policies, and the quantities the paper reports.

3.1. Time, causes and actions

Definition 1 (Time, hidden cause, model variants). Time is discrete, $t = 0, 1, \dots, T$, with horizon $T \geq 1$; decisions are taken at $t = 0, \dots, T - 1$ and the incident is under way from $t = 0$ (the alarm time). The hidden cause M is drawn once at $t = 0$ and does not change. In the *two-state model* $M \in \mathcal{M}_2 = \{K, F\}$ (compromise, fault) and the belief is the scalar $p_t = P(M = K \mid \mathcal{I}_t) \in [0, 1]$, where $\mathcal{I}_t$ is the history of actions and observations before the decision at t . In the *three-state model* $M \in \mathcal{M}_3 = \{K, F, B\}$, where B is a benign market shock (no compromise, no fault), and the belief is the point $\pi_t = (\pi_t(K), \pi_t(F), \pi_t(B))$ of the 2-simplex $\mathcal{S}_2$. The prior is p_0 (resp. π_0).

Definition 2 (Actions). At each decision time the operator chooses one action from

$$\mathcal{A} = \{\text{monitor, throttle, verify, kill, leave}\}.$$

monitor and **throttle** are *postures* held for one period, and **verifying** is the posture held while verification runs; **verify** is the verification action of Paper 2 (Definition 2 there), available at most once per episode, which holds the *verifying* posture for $\ell_v \geq 1$ periods and returns its outcome at the end of them (no decision is taken during these periods); **kill** (contain) and **leave** (stand down) are *terminal* and end the episode. At $t = T$ only **kill** and **leave** are available. A *verification flag* $v_t \in \{0, 1\}$ records whether verification has been used; the decision state is (t, p_t, v_t) (resp. (t, π_t, v_t)).

The ladder has three postures and two terminal actions. **monitor** lets trading continue while the operator watches; **throttle** limits the flow (for instance by rate limits or reduced order sizes) at a disruption cost, reducing harm; both are held for one period. **verify** queries a channel that order flow does not carry and holds the third posture, **verifying**, for the ℓ_v periods the check takes, during which no decision is taken; **kill** contains (suspends the session, pulls the orders); **leave** stands down. Leave is needed both so that the kill-switch baselines can stand down as well as kill, and for the exact embedding of Paper 2's rule (Proposition 3).

3.2. Observations and the discriminability factor

Definition 3 (Order-flow observations and separation). In each period spent in a posture $x \in \{\text{monitor, throttle}\}$ the operator observes an order-flow statistic $y \in \mathbb{R}$; in a period of the **verifying** posture it observes one with the law of the **monitor** posture if $\omega = 1$ and none if $\omega = 0$ (Definition 4). In the two-state model,

$$y \mid (M = K, x) \sim \mathcal{N}\left(+\frac{1}{2}d_x, 1\right), \quad y \mid (M = F, x) \sim \mathcal{N}\left(-\frac{1}{2}d_x, 1\right), \quad d_{\text{monitor}} = d, \quad d_{\text{throttle}} = \kappa d,$$

with per-period separation $d \geq 0$ and *discriminability factor* $\kappa \geq 0$ (the restriction $\kappa \leq 1$ is imposed by Assumption 4, not by this definition). In the three-state model, $y \in \mathbb{R}^k$ ($k \in \{1, 2\}$) with $y \mid (M, x) \sim \mathcal{N}(g_x \mu_M, I_k)$, $g_{\text{monitor}} = 1$, $g_{\text{throttle}} = \kappa$, for fixed means $\mu_K, \mu_F, \mu_B \in \mathbb{R}^k$; the pairwise per-period separations are $d_{MM'} = \|\mu_M - \mu_{M'}\|$ under monitoring and $\kappa d_{MM'}$ under throttling. The two-state model is the three-state model restricted to $\{K, F\}$ with $\|\mu_K - \mu_F\| = d$: for $d > 0$ the statistic $u^\top (y - g_x(\mu_K + \mu_F)/2)$, $u = (\mu_K - \mu_F)/d$, is sufficient for $\{K, F\}$ and has the two-state laws above.

The factor κ is the share of the per-period separation that throttling keeps: in Paper 1’s linear model it is the ratio $\kappa_W = d_W(\Theta)/d_W(\emptyset)$ of whitened separations with and without the throttle (Definition 11).

The separation d is the distance between the two means in units of the noise. Throttling moves the two means towards each other by the factor κ . At $\kappa = 1$ throttling costs no evidence, and at $\kappa = 0$ a throttled period carries no evidence at all.

Definition 4 (Verification outcome). The outcome $Z \in \{+, -\}$ of **verify** has the law of Paper 2, Definition 2: $P(Z = + \mid K) = s$, $P(Z = - \mid F) = t_F$ (Paper 2 writes the specificity t). In the three-state model $P(Z = - \mid B) = t_B$, with $t_B = t_F$ unless stated otherwise (verification reads authorization, which is valid under both F and B). During the ℓ_v periods of the **verifying** posture the operator also observes the order-flow statistics of the **monitor** posture if $\omega = 1$ and none if $\omega = 0$, and harm accrues at the harm rate of the cause, r_K under K and r_F under F (r_B under B), as in the **monitor** posture (Definition 6); $\omega = 0$ is the analogue of Paper 2, Assumption 4 (no further order-flow evidence during the delay).

Definition 5 (Belief update). After an order-flow statistic y observed in posture x , the two-state belief is updated by Bayes’ rule (in the **verifying** posture with $\omega = 1$, $d_x = d$),

$$\text{logit } p' = \text{logit } p + d_x y,$$

(the log-likelihood ratio of the two Gaussians of Definition 3), and after a verification outcome z by $p' = p^z$ with p^+, p^- as in Paper 2, Definition 2. In the three-state model, $\pi'(M) \propto \pi(M) \phi_k(y - g_x \mu_M)$ after y and $\pi'(M) \propto \pi(M) P(Z = z \mid M)$ after z , where ϕ_k is the standard normal density on $\mathbb{R}^k$.

3.3. Costs and optimal value

Definition 6 (Costs). The terminal costs form the loss table (C_K, C_F, H_K, H_F) , extended by (C_B, H_B) in the three-state model: **kill** (containing) at any t costs C_M under cause M and ends the episode; **leave** (standing down) at any t costs H_M and ends it. Write $a = H_K - C_K$ (what containing saves under compromise) and $b = C_F - H_F$ (what containing costs needlessly under a fault). Each period spent in a posture adds a stage cost: **monitor** costs r_M (the harm accrued in the period under cause M), **throttle** costs $\delta + \rho r_M$, with disruption cost $\delta \geq 0$ and *harm factor* $\rho \in [0, 1]$; **verify** costs c_v once plus r_M for each of its ℓ_v periods in the **verifying** posture (harm is not reduced while

verifying). The baseline harm rates are $r_K \geq 0$ and $r_F = r_B = 0$. Stage costs are additional to the terminal cost, whichever terminal action ends the episode, and there is no discounting.

In the loss table, a is what containing saves under a compromise and b what containing costs needlessly under a fault; they fix the kill/leave indifference belief $q^* = b/(a + b)$ of Assumption 1, on which the terminal choice turns. The throttle trades a certain disruption cost δ against a reduction of expected harm from pr_K to ρpr_K , and it pays for this reduction with evidence, through κ . Whether that trade is worth making is what the value of throttling of Definition 10 measures.

Definition 7 (Policies, value, Bellman recursion). A policy maps each decision state to an action. Its *expected cost* $J^\pi(p_0)$ is the expected sum of stage and terminal costs from prior p_0 with $v_0 = 0$. In the two-state model the optimal value $V_t(p, v)$ satisfies $V_T(p, v) = R(p) = \min\{pC_K + (1 - p)C_F, pH_K + (1 - p)H_F\}$ and, for $t < T$,

$$V_t(p, v) = \min_{a \in \mathcal{A}_t(v)} Q_t(p, v, a),$$

$$\begin{aligned} Q_t(p, v, \text{kill}) &= pC_K + (1 - p)C_F, & Q_t(p, v, \text{leave}) &= pH_K + (1 - p)H_F, \\ Q_t(p, v, \text{monitor}) &= pr_K + \mathbb{E}_y[V_{t+1}(p', v)], \\ Q_t(p, v, \text{throttle}) &= \delta + \rho pr_K + \mathbb{E}_y[V_{t+1}(p', v)], \\ Q_t(p, 0, \text{verify}) &= c_v + \ell_v pr_K + \mathbb{E}[V_{t+\ell_v}(p'', 1)], \end{aligned}$$

where p' is the update of Definition 5 in the posture taken, the expectation is over y under the predictive mixture $p\mathcal{N}(\frac{1}{2}d_x, 1) + (1 - p)\mathcal{N}(-\frac{1}{2}d_x, 1)$, p'' is the belief after the verification outcome (and, if $\omega = 1$, the ℓ_v observations of the **verifying** posture), $\mathcal{A}_t(1)$ excludes **verify**, and **verify** is excluded when $t + \ell_v > T$. The three-state recursion is the same with p replaced by $\pi \in \mathcal{S}_2$ and every $pX_K + (1 - p)X_F$ by $\sum_M \pi(M)X_M$. Ties are broken towards the lower rung of the order **leave**, **monitor**, **verify**, **throttle**, **kill**.

3.4. Numerical solution

Definition 8 (Belief grid). The recursion of Definition 7 is solved by backward induction on a grid: in the two-state model the uniform grid $G_N = \{i/N : i = 0, \dots, N\}$, with V_{t+1} evaluated off the grid by piecewise-linear interpolation and $\mathbb{E}_y$ computed by a Bayes-consistent quadrature rule of order n_q (nodes and weights independent of p , Proposition 1(iii)); in the three-state

model the triangular grid $G_N^{(2)} = \{\pi \in \mathcal{S}_2 : N\pi \in \mathbb{Z}^3\}$ with barycentric linear interpolation and $\mathbb{E}_y$ computed by Gauss–Hermite quadrature of order n_q in each Gaussian component (the *per-component Gauss–Hermite rule*, also a stated variant in the two-state model). The reported value is the grid value; a solution is *grid-converged at tolerance ϵ* if doubling N and n_q changes every reported value (the optimal value $V_0(p_0, 0)$, the values of the restricted recursions of Definition 9 and the expected cost of every evaluated policy) by at most ϵ at every reported point. No reinforcement learning or function approximation is used.

The two-state recursion is solved with a Bayes-consistent quadrature rule, under which the grid problem is itself an exact Bayes recursion for a discrete observation. The rule is one Gauss–Legendre rule of order n_q in y , with nodes symmetric about the midpoint of the two means and shared by both causes, and the two likelihoods as factors, so that its nodes and weights do not depend on the belief; concavity of the grid values (Proposition 1(iii)) is then a checkable property of the solver’s output, used as a test of the implementation. The three-state recursion uses the per-component Gauss–Hermite rule of Definition 8. The solver is written in NumPy; the code, with the version used for every reported computation, is to be made available in an archived repository.

3.5. Baselines and reported quantities

Definition 9 (Baseline policies). (i) *Fixed-threshold kill-switch* KS_θ : **monitor** while $p_t < \theta$, **kill** as soon as $p_t \geq \theta$, and at $t = T$ the better terminal action at p_T . (ii) *Best Bayesian kill-switch* KS^* : the optimal policy of the recursion of Definition 7 with the action set restricted to $\{\text{monitor}, \text{kill}, \text{leave}\}$ (an optimal stopping rule on the belief, with time-dependent thresholds). Its value is V_t^{KS} ; by construction $V_0^{\text{KS}} \leq J^{\text{KS}_\theta}$ for every θ . (iii) *Kill-switch with verification* KSV^* and *kill-switch with throttle* KST^* : the same with **verify**, respectively **throttle**, added to the restricted set; their values are V_t^{KSV} and V_t^{KST} . (iv) *Paper 2’s one-shot rule* P2 : at $t = 0$, with belief p_0 , **verify** if p_0 lies in Paper 2’s verification region S (Paper 2, Definition 3, with Paper 2’s delay harm, the harm rate times Paper 2’s delay, read as $\ell_v r_K$), then take the better terminal action at the posterior; otherwise take the better terminal action at p_0 . No order flow is used. (v) *Observability-blind throttle policy* NB : at each decision state (t, p_t, v_t) it takes the action of the optimal policy of Definition 7 computed with $\kappa = 1$ (tie-breaking of Definition 7), where p_t is the Bayes belief of the true model: in throttled periods it updates with

the true separation κd (Definition 5). NB is thus a correct filter with a misspecified planner; the belief update with the misspecified separation d is not used. Its expected cost J^{NB} is evaluated in the true model by the recursion of Definition 7 with the minimum replaced by NB's action.

The best Bayesian kill-switch KS^* is the strongest kill-switch the model allows: it chooses optimally when to stop, and on which side, from the same belief as the graded policy, but may not throttle or verify. The restricted policies KSV^* and KST^* add one rung each, which splits the gain of the full ladder into a verify share and a throttle share. The observability-blind policy NB is the operator who plans as if throttling kept full discriminability but updates the belief correctly: a correct filter with a misspecified planner, whose cost is evaluated in the true model.

Definition 10 (Operating characteristics, gains, critical discriminability factor). For a policy π and prior p_0 : the *false-kill rate* $\text{FPR}^\pi = P^\pi(\text{kill} \mid F)$; the *missed-containment rate* $\text{FNR}^\pi = P^\pi(\text{no kill} \mid K)$; the *expected time to a terminal action*; and the *expected throttled time*. The *graded gain* is $\Gamma(p_0) = V_0^{\text{KS}}(p_0, 0) - V_0(p_0, 0) \geq 0$, with the decomposition into the verify share $V_0^{\text{KS}} - V_0^{\text{KSV}}$ and the throttle share $V_0^{\text{KSV}} - V_0^{\text{KST}}$. The *value of throttling* at discriminability factor κ is $G(\kappa) = V_0^{\text{KSV}}(p_0, 0) - V_0(p_0, 0; \kappa) \geq 0$, where V^{KSV} (Definition 9(iii)) is the optimal value with every action but throttle (it does not depend on κ). The *critical discriminability factor* is $\kappa^* = \inf\{\kappa \in [0, 1] : G(\kappa) > 0\}$ ($\kappa^* = 1$ if the set is empty).

Every comparison is in expected cost at the prior, and every policy is reported with its false-kill and missed-containment rates, the expected time to a terminal action and the expected throttled time beside its cost. *Dominance* of a policy over another means strictly lower expected cost at the prior, $\Gamma(p_0) > 0$ for the graded gain; it is claimed only on a stated region of parameters, never for all of them.

3.6. Assumptions

Assumption 1 (Cost orderings). $H_K > C_K$ and $C_F > H_F$, that is $a > 0$ and $b > 0$ (Paper 2, Assumption 1); in the three-state model also $C_B > H_B$. In the two-state model the *kill/leave indifference belief* is $q^* = b/(a + b) \in (0, 1)$, the unique belief at which kill and leave have the same expected terminal cost: $(pH_K + (1 - p)H_F) - (pC_K + (1 - p)C_F) = (a + b)(p - q^*)$. Stage costs satisfy $r_K \geq 0$, $\delta \geq 0$, $c_v \geq 0$, $\rho \in [0, 1]$.

Assumption 2 (Conditionally i.i.d. Gaussian order flow). Given M and the postures taken, the per-period order-flow statistics are indepen-

dent, with the laws of Definition 3; the separation d does not change with time. (A time-varying separation d_t is a stated variant, not the baseline.)

Assumption 3 (Separate channel). The verification outcome Z is conditionally independent of all order-flow statistics given M (Paper 2, Assumption 2), and $s + t_F > 1$.

Assumption 4 (Throttle effects). $0 \leq \kappa \leq 1$: throttling never makes the causes easier to tell apart. This is a restriction of this paper’s model, not a consequence of Paper 1’s: Proposition 4(iv) gives instances with $\kappa_W > 1$. ρ and κ are treated as independent parameters; the linear-model relations between them (Proposition 4) are stated where used.

Assumption 5 (Known model; non-strategic cause). All parameters (d , κ , ρ , δ , s , t_F , c_v , ℓ_v , the loss table, the prior) are known to the operator, and the cause does not react to the operator’s actions (a strategic attacker is outside this paper’s scope).

Assumption 2 treats the per-period statistics as independent with a constant separation. In the linear-Gaussian model of Paper 1 the separation can grow with time through state feedback; a time-varying separation d_t is a stated variant. Assumption 4 excludes throttles that make the causes easier to tell apart; Section 3.7 shows that it is a restriction of the model, not a consequence of Paper 1’s.

Remark 1 (Equivalent-Gaussian calibration). Paper 1’s ABIDES constructions were never shown to be Gaussian. A separation backed out of a Paper 1 accuracy through the Gaussian benchmark $\text{Acc}^* = \Phi(d_W/2)$ is an *equivalent-Gaussian calibration heuristic*, and it is a separation d_W for the classifier’s whole observation window, not a per-period separation. Under Assumption 2, n_W periods of monitoring give separation $d\sqrt{n_W}$, so the heuristic per-period value is $d = d_W/\sqrt{n_W}$, where n_W is the number of decision periods the window is taken to span (a modelling choice, stated wherever used). Every result evaluated at a calibrated d is also reported over a direct sweep of d .

3.7. Where $\kappa < 1$ comes from

The discriminability factor is a parameter of the POMDP. To see what it stands for, we return to the linear-Gaussian model of Paper 1 (Mejrissi, 2026b), in which an incident’s injection enters the observation of the book and, under state feedback, its latent state, and model two throttles in it.

Definition 11 (Throttles in Paper 1’s linear-Gaussian model). Take Paper 1’s output-level model (Definition D2) or state-level model (Def-

inition D9) for a pair of mechanism instances (K, θ_K, b_K) , (F, θ_F, b_F) (Definition D1), under Paper 1's assumptions (A1)–(A4) and (A6), with onset τ and $\Delta_0 = \theta_K b_K - \theta_F b_F$. A *throttle schedule* is a set Θ of periods, the throttled periods; $\Theta = \emptyset$ is monitoring throughout. (i) A *scaling throttle* with factor $r \in [0, 1]$ replaces the injection $\theta_M b_M$ by $r \theta_M b_M$ in every throttled period, wherever the injection enters (the observation and, under D9, the state recursion). (ii) A *feedback-suppressing throttle* keeps the injection in the observation and removes it from the state recursion in throttled periods: under D9, $x_{t+1} = Ax_t + w_t$ for $t \in \Theta$ (the throttled periods follow D2). For a finite window W (Paper 1, Definition D3) let $d_W(\Theta)$ be the whitened mean separation of $Y_W = (y_t)_{t \in W}$ under schedule Θ (Paper 1, Definition D5). When $d_W(\emptyset) > 0$, the *window discriminability factor* is $\kappa_W(\Theta) = d_W(\Theta)/d_W(\emptyset)$. Under Assumption 2, a window of n periods in one posture has separation $d\sqrt{n}$ under monitoring and $\kappa d\sqrt{n}$ under throttling, so $\kappa_W = \kappa$: κ_W is the quantity of Paper 1's model that this paper's κ stands for.

Proposition 4 (Where $\kappa < 1$ comes from). In the setting of Definition 11, let W be a finite window and $t_W = \max W$. (i) (*Scaling throttle.*) If $\Theta \supseteq \{\tau, \dots, t_W\}$, the mean difference of Y_W between K and F under Θ is r times the one under monitoring, in D2 and in D9; hence $\kappa_W(\Theta) = r$ whenever $d_W(\emptyset) > 0$. Under D2 it suffices that $\Theta \supseteq W \cap \{t \geq \tau\}$. (ii) (*Feedback-suppressing throttle on an output-confounded pair.*) Suppose the model is D9 and $C\Delta_0 = 0$. If $\Theta \supseteq \{\tau, \dots, t_W\}$, then Y_W has the same law under K and F , so $d_W(\Theta) = 0$. Under monitoring, $d_W(\emptyset) > 0$ if and only if $C \sum_{j=1}^k A^j \Delta_0 \neq 0$ for some $k \geq 1$ with $\tau + k \in W$; if $W \cap \{t \geq \tau\} = \{\tau, \dots, \tau + H - 1\}$, this holds if and only if $CA^j \Delta_0 \neq 0$ for some $1 \leq j \leq H - 1$. When it holds, $\kappa_W(\Theta) = 0$. (iii) (*Late start.*) If the throttle starts after the onset, $\Theta = \{t_0, t_0 + 1, \dots\}$ with $t_0 > \tau$, (i) and (ii) can fail. Let $D_{t_0} = \sum_{j=1}^{t_0 - \tau} A^j \Delta_0$. Under D9, for $t \geq t_0$, the mean difference of y_t is r times the one under monitoring plus $(1 - r)CA^{t-t_0}D_{t_0}$ for the scaling throttle, and $CA^{t-t_0}D_{t_0} + C\Delta_0$ for the feedback-suppressing throttle. The residual term $CA^{t-t_0}D_{t_0}$, the memory of the unthrottled feedback before t_0 , is non-zero at $t = t_0$ for some (A, C, Δ_0) with $C\Delta_0 = 0$; for those, the feedback-suppressing throttle has $d_W(\Theta) > 0$ for every window $W \subseteq \Theta$ that contains t_0 , and the scaling throttle with $r < 1$ has $\kappa_W(\Theta) \neq r$ for $W = \{t_0\}$. (iv) (*Other pairs.*) If $C\Delta_0 \neq 0$, the factor κ_W of a feedback-suppressing throttle with $\Theta \supseteq \{\tau, \dots, t_W\}$ is not signed by the model: there are instances with $\kappa_W < 1$, $\kappa_W = 1$ and $\kappa_W > 1$: in the scalar model D9 with $C = 1$, $Q = 0$, $\Delta_0 \neq 0$ and $W = \{\tau, \tau + 1\}$,

$\kappa_W^2 = 2/(1 + (1 + A)^2)$, so $A = 1$ gives $\kappa_W = \sqrt{2/5}$, $A = 0$ gives $\kappa_W = 1$ and $A = -1$ gives $\kappa_W = \sqrt{2}$. In particular Assumption 4 ($\kappa \leq 1$) is a restriction of this paper’s model, not a consequence of Paper 1’s.

Proposition 4 is proved directly from Paper 1’s model definitions; it uses neither of Paper 1’s theorems. Part (i) ties the two effects of a scaling throttle together: in Paper 1’s model a throttle that scales the injection by r has $\kappa = r$, so the harm factor and the discriminability factor move together. Part (ii) is the sharpest case. For a pair confounded at the output level, $C\Delta_0 = 0$, all the discriminability of monitoring comes from the feedback of the injection into the state, and a throttle that removes that feedback from the onset leaves none: $\kappa = 0$. The condition under which monitoring separates such a pair over a finite window is the negation of the condition of Paper 1’s Theorem 3 (the finite-horizon version of its Theorem 2); this is the sense in which Paper 1’s Theorem 2 is the mechanism of part (ii). Part (iii) qualifies part (ii): an operator throttles after the alarm, not at the onset, and the feedback accumulated before the throttle starts leaves a residual term that decays according to the dynamics. The value $\kappa = 0$ is exact only from the onset, and otherwise holds after the transient. Part (iv) shows that for pairs that are not confounded at the output level the sign of $\kappa - 1$ is not determined by the model, which is why the POMDP sweeps κ over $[0, 1]$ and states $\kappa \leq 1$ as Assumption 4. In the POMDP, ρ and κ are treated as independent parameters; the diagonal $\kappa = r$ of part (i) is a distinguished case.

4. Structure of the optimal policy

This section states what can be proved about the optimal policy for every parameter value. Sections 4.1–4.4 give the structure of the discrete model; Section 4.5 shows that a throttle that blinds too much is never used; Section 4.6 gives, in a diffusion model, the exact condition under which throttling is worth its loss of evidence. The proofs are given in Appendix A, which also states the status of each. Section 5 then asks what cannot be proved in general: whether the gains these results allow are strictly positive and material on the calibrated region.

Lemma 1, Proposition 1, Theorem 1 and Proposition 2 are classical results of the theory of partially observed Markov decision processes, specialised to this model; none of them is claimed as new. Concavity of the finite-horizon value in the belief goes back to Smallwood and Sondik (1973), who

showed that it is piecewise linear and concave. That a more informative observation is worth more to a decision maker is Blackwell’s comparison of experiments (Blackwell, 1953), and conditions under which the value of a partially observed problem and its optimal policy are monotone in the belief are given by Lovejoy (1987); structural results of both kinds for POMDPs and controlled sensing are collected by Krishnamurthy (2025). We state and prove them for the model of Section 3 so that the assumptions each one needs here are explicit, and because the numerical study uses them as checks of the solver (Section 5.4).

4.1. Concavity

Every result of this section rests on one operator: the expected value of a function of the Bayes posterior after one observation.

Lemma 1 (Bayes operator: concavity, continuity, garbling). Let ν be a σ -finite measure on an observation space O and $\lambda_K, \lambda_F \geq 0$ be ν -integrable. For a bounded Borel $f : [0, 1] \rightarrow \mathbb{R}$ and $p \in [0, 1]$ put

$$(T_\lambda f)(p) = \int_O \Lambda_p(o) f(q_p(o)) \nu(do), \quad \Lambda_p = p \lambda_K + (1-p) \lambda_F, \quad q_p = \frac{p \lambda_K}{\Lambda_p},$$

with the integrand read as zero where $\Lambda_p(o) = 0$. When λ_K, λ_F are probability densities, $(T_\lambda f)(p) = \mathbb{E}[f(p')]$ for the Bayes posterior p' after one observation drawn from the predictive law $\Lambda_p \nu$. (i) If f is concave, $T_\lambda f$ is concave on $[0, 1]$. (ii) If f is continuous, $T_\lambda f$ is continuous on $[0, 1]$. (iii) (*Garbling.*) Let λ'_K, λ'_F be probability densities with respect to a σ -finite ν' on O' , and let $\mathcal{G}(o | o') \geq 0$ be jointly measurable with $\int_O \mathcal{G}(o | o') \nu(do) = 1$ for every o' . If $\lambda_M(o) = \int_{O'} \mathcal{G}(o | o') \lambda'_M(o') \nu'(do')$ for $M \in \{K, F\}$ (the same kernel for both causes), then $T_\lambda f \geq T_{\lambda'} f$ on $[0, 1]$ for every concave continuous f . (iv) (*Gaussian garbling.*) For $0 \leq r < 1$ and $k \geq 1$, the kernel $\mathcal{G}(y | y') = \phi_k((y - ry')/\sigma_r)/\sigma_r^k$, $\sigma_r = \sqrt{1 - r^2}$, maps $\mathcal{N}(m, I_k)$ to $\mathcal{N}(rm, I_k)$ for every $m \in \mathbb{R}^k$. (i)–(iii) hold in the three-state model with p replaced by $\pi \in \mathcal{S}_2$, $\Lambda_\pi = \sum_M \pi(M) \lambda_M$ and the posterior $\pi_o(M) = \pi(M) \lambda_M(o)/\Lambda_\pi(o)$.

Part (iii) is Blackwell’s ordering (Blackwell, 1953) in the form used here: an observation obtained from another by a noise kernel common to both causes is worth less to an operator whose continuation value is concave. Part (iv) shows that throttling is such a garbling: the Gaussian law of Definition 3 at separation κd is a garbling of the law at separation d .

Proposition 1 (Concavity). Under Definitions 1–7 and Assumptions 2–3, for every $t \in \{0, \dots, T\}$ and $v \in \{0, 1\}$: (i) in the two-state model $V_t(\cdot, v)$

and every $Q_t(\cdot, v, a)$ are concave and continuous on $[0, 1]$; (ii) in the three-state model $V_t(\cdot, v)$ and every $Q_t(\cdot, v, a)$ are concave and continuous on $\mathcal{S}_2$; (iii) (*grid, two-state model*) if in Definition 8 every expectation $\mathbb{E}[g(p')]$ is replaced by a *Bayes-consistent* rule $\sum_j \omega_j \Lambda_p(o_j) g(q_p(o_j))$ with nodes o_j and weights $\omega_j \geq 0$ that do not depend on p (the operator T_λ of Lemma 1 for the discrete measure $\sum_j \omega_j \delta_{o_j}$), then for every t, v the grid values $(\tilde{V}_t(i/N, v))_{i=0}^N$ form a concave sequence and their piecewise-linear interpolant is concave on $[0, 1]$. No grid concavity is claimed for the per-component Gauss–Hermite rule of Definition 8, which is not Bayes-consistent, nor for the triangular grid of the three-state model, on which piecewise-linear interpolation of concave data need not be concave.

Part (iii) is used in the numerical study as a test of the solver: with the Bayes-consistent quadrature rule, grid values that fail to be concave in the belief reveal an implementation error, not a property of the model.

4.2. Monotonicity in discriminability and the critical factor

Theorem 1 (Monotonicity in discriminability; a single critical factor). Under Definitions 1–7, 10 and Assumptions 2–4, with all other parameters fixed: (i) for $0 \leq \kappa < \kappa' \leq 1$, $V_t(\cdot, v; \kappa) \geq V_t(\cdot, v; \kappa')$ for every t and v , in the two-state and in the three-state model; (ii) in the two-state model, for $0 \leq d < d'$, $V_t(\cdot, v; d) \geq V_t(\cdot, v; d')$ for every t and v (the monitoring separation is d and the throttling separation κd in both); (iii) at every prior p_0 , the value of throttling $G(\kappa)$ is non-negative and non-decreasing on $[0, 1]$, so $\{\kappa \in [0, 1] : G(\kappa) > 0\}$ is empty, $(\kappa^*, 1]$ or $[\kappa^*, 1]$; (iv) for $\kappa < \kappa^*$, $G(\kappa) = 0$: an optimal policy of the recursion without **throttle** is optimal from p_0 , so an optimal operator need never throttle; for $\kappa > \kappa^*$, $G(\kappa) > 0$ and every policy that is optimal from p_0 throttles with positive probability. (iii) and (iv) hold in the three-state model with p_0 replaced by π_0 .

Theorem 1(iii)–(iv) is what makes the critical discriminability factor κ^* (Definition 10) a single threshold: the value of throttling $G(\kappa)$ cannot switch on, off and on again as κ grows. Below κ^* the throttle has no value, and a policy that never throttles is optimal; above it, every optimal policy throttles with positive probability. The statement is weak in one direction on purpose: the tie-broken policy of Definition 7 may still throttle below κ^* at a belief where throttling ties with killing, at no change in cost, so an optimal operator *need* never throttle there. Theorem 1 does not locate κ^* . Its lower end is fixed by Proposition 6 (Section 4.5): when throttling has a disruption cost,

$\kappa^* > 0$. Whether $\kappa^* < 1$ on the calibrated region, that is whether the throttle has value there for a discriminability factor high enough, rather than never ($\kappa^* = 1$), is claim 3, which the numerical study tests; Proposition 5 (Section 4.6) gives the location of κ^* in closed form in a diffusion model.

4.3. Kill and leave regions

Proposition 2 (Kill and leave regions). Under Definitions 1–7 and Assumptions 1–3, in the two-state model let $q^* = b/(a + b)$ be the kill/leave indifference belief (Assumption 1) and, for $t \leq T$ and $v \in \{0, 1\}$, let $\mathcal{K}_t(v) = \{p : Q_t(p, v, \text{kill}) = V_t(p, v)\}$ and $\mathcal{L}_t(v) = \{p : Q_t(p, v, \text{leave}) = V_t(p, v)\}$ be the sets where kill, respectively leave, is optimal. Then: (i) $V_t(1, v) = C_K$ and $V_t(0, v) = H_F$; (ii) $\mathcal{K}_t(v) = [p_t^K(v), 1]$ and $\mathcal{L}_t(v) = [0, p_t^L(v)]$ are closed intervals with $p_t^L(v) \leq q^* \leq p_t^K(v)$, and $\mathcal{K}_t(v) \cap \mathcal{L}_t(v) \subseteq \{q^*\}$; (iii) under the tie-breaking of Definition 7 the optimal policy takes **leave** exactly on $\mathcal{L}_t(v)$, and takes **kill** on the set where kill is the unique optimal action, an interval that is open relative to $[0, 1]$ and contained in $\mathcal{K}_t(v)$; if $r_K > 0$ and $\delta + \rho r_K > 0$, that interval contains 1. (iv) In the three-state model (with $C_B > H_B$, Assumption 1), the set where **kill** is optimal is convex and contains the vertex e_K , and the set where **leave** is optimal is convex and contains the edge $[e_F, e_B]$. Whether **monitor**, **throttle** and **verify** occupy ordered bands between $p_t^L(v)$ and $p_t^K(v)$ is not claimed.

The operator kills at high beliefs and leaves at low ones, and the two regions are separated by the kill/leave indifference belief q^* of Assumption 1. Between them lie the beliefs at which it pays to keep watching, to throttle or to verify. Proposition 2 does not say how **monitor**, **throttle** and **verify** are arranged in that middle region; the optimal-action maps of the numerical study show it.

4.4. Embedding of the one-shot rule

Proposition 3 (Embedding of Paper 2). Under Definitions 1–7, 9 and Assumptions 1–3, in the two-state model let R and R_v be Paper 2’s passive and verify-first risks (Paper 2, Definitions 1 and 3) with Paper 2’s delay harm (the harm rate times Paper 2’s delay) read as $\ell_v r_K$, and $S = \{p \in [0, 1] : R_v(p) < R(p)\}$ Paper 2’s verification region (Definition 3 there). Suppose $T \geq \ell_v$. (i) For every $d \geq 0$, the one-shot rule P2 of Definition 9(iv) has expected cost $J^{\text{P2}}(p) = \min\{R(p), R_v(p)\}$. (ii) (*Order flow uninformative.*) If $d = 0$, then, whatever ω , $V_t(p, 1) = R(p)$ for every $t \leq T$, and $V_t(p, 0) = \min\{R(p), R_v(p)\}$ if $t + \ell_v \leq T$ and $V_t(p, 0) = R(p)$ otherwise; in particular

$V_0(p, 0) = J^{P2}(p)$. (iii) If $d = 0$, $r_K > 0$ and $\delta + \rho r_K > 0$, the optimal policy of Definition 7 (with its tie-breaking) from p is P2: at $t = 0$ it takes **verify** if $p \in S$, **leave** if $p \notin S$ and $p \leq q^*$, **kill** if $p \notin S$ and $p > q^*$, except at beliefs $p > q^*$ with $R_v(p) = R(p)$, where **verify** and **kill** are both optimal and Definition 7 breaks the tie towards **verify** while Paper 2 breaks it towards containing; after **verify**, at $t = \ell_v$ it takes the better terminal action at the posterior (**leave** on a tie). If $r_K = 0$ or $\delta + \rho r_K = 0$, (ii) still holds, but the tie-broken policy may **monitor** or **throttle** before acting, at the same expected cost. (iv) (*Sequential weakly beats one-shot.*) For every $d \geq 0$, $V_0(p, 0; d) \leq J^{P2}(p)$. If $T < \ell_v$, **verify** is never available (Definition 7); in particular, at $T = 1$ the embedding requires $\ell_v = 1$.

When order flow carries no information, the sequential problem collapses to Paper 2's one-shot choice between leaving, verifying and containing, and the optimal value equals the cost of Paper 2's rule. For every d , the sequential optimum is at least as good as the one-shot rule, by part (iv). That it is strictly better when order flow is informative and the horizon allows more than one decision is claim 2, a corollary of Proposition 3 rather than a headline result: the numerical study reports its magnitude, and whether that magnitude is material, on the calibrated region.

4.5. A blinding throttle is never used

A throttle with $\kappa = 0$ removes all evidence from the periods it is in force. It still reduces harm, and one might expect it to be worth using when harm is high and disruption cheap. The next result shows that it never is.

Proposition 6 (A blinding throttle is never used). Under Definitions 1–7, 9, 10, Assumptions 1–3 and Proposition 1 (continuity; used in (iii) for the value V^{KSV} of a restricted recursion of Definition 9), in the two-state and in the three-state model, with all parameters but κ fixed: (i) $V_t(\cdot, v) \leq V_{t+1}(\cdot, v)$ for $0 \leq t < T$ and $v \in \{0, 1\}$, and the same holds for the values V_t^{KS} , V_t^{KSV} , V_t^{KST} of the restricted recursions of Definition 9; (ii) at $\kappa = 0$, $Q_t(p, v, \text{throttle}) \geq V_t(p, v) + \delta + \rho p r_K$ for every $t < T$, p and v (in the three-state model with $\pi(K)$ in place of p), and $V_t(\cdot, v; 0) = V_t^{\text{KSV}}(\cdot, v)$; in particular $G(0) = 0$ at every prior; (iii) if $\delta > 0$, there is $\kappa_0 \in (0, 1]$, which does not depend on the prior, such that for every $\kappa \in [0, \kappa_0]$ **throttle** is not optimal at any decision state and $G(\kappa) = 0$; hence $\kappa^* \geq \kappa_0 > 0$.

A throttle with $\kappa = 0$ is a paid pause: it leaves the belief where it was, costs the disruption and the reduced harm of the period, and shortens the time left to decide, which by part (i) can only raise the cost to go. So the

fully blinding throttle is never strictly better than the best action without it, whatever the harm rate, and $G(0) = 0$ at every prior. With a disruption cost, part (iii) extends this from $\kappa = 0$ to a whole interval of factors, so κ^* is strictly positive. In the design of the numerical study the disruption cost is positive at every point, so a computed $\kappa^* = 0$ would be an error of the solver, not evidence (Section 5.3). Part (iii) applies to the value V^{KSV} of a restricted recursion of Definition 9 a continuity property that Proposition 1 states for the recursion of Definition 7; the proof in Appendix A states this dependency and a proposed closure, which has not yet been independently verified.

4.6. A cost-per-information law in a diffusion model

Theorem 1 and Proposition 6 bound κ^* but do not locate it. In the discrete model of Section 3 its location is computed (Section 5). In a continuous-time version of the two-state model, without a horizon, without discounting and without **verify**, it has a closed form.

Definition 12 (Diffusion model). The two-state model in continuous time $t \geq 0$, with no horizon, no discounting and no **verify**. A *policy* is given on a filtered probability space carrying M (with $P(M = K) = p_0$) and a standard Brownian motion B independent of M (weak formulation): a posture process $x_t \in \{\text{monitor}, \text{throttle}\}$ and a stopping time $\tau < \infty$ a.s., both with respect to the filtration $(\mathcal{F}_t^Y)$ of the order flow

$$Y_t = \int_0^t \kappa_{x_s} d\theta_M ds + B_t, \quad \theta_K = +\frac{1}{2}, \theta_F = -\frac{1}{2}, \quad \kappa_{\text{monitor}} = 1, \kappa_{\text{throttle}} = \kappa \in [0, 1],$$

with x progressively measurable; at τ the better terminal action at p_τ is taken, where $p_t = P(M = K \mid \mathcal{F}_t^Y)$. Its expected cost is $J(p_0) = \mathbb{E}\left[\int_0^\tau c_{x_t}(p_t) dt + g(p_\tau)\right]$, with running costs $c_{\text{monitor}}(p) = p r_K$, $c_{\text{throttle}}(p) = \delta + \rho p r_K$ and $g(p) = \min\{p C_K + (1-p) C_F, p H_K + (1-p) H_F\}$. $V^D(p_0)$ is the infimum of J over policies and $V^{D,\text{KS}}(p_0)$ the infimum over policies with $x \equiv \text{monitor}$; $G^D(\kappa) = V^{D,\text{KS}} - V^D$ and $\kappa_D^* = \inf\{\kappa \in [0, 1] : G^D(\kappa) > 0\}$ ($\kappa_D^* = 1$ if the set is empty). The *throttle-free continuation interval* is $\mathcal{C}_0 = \{p : V^{D,\text{KS}}(p) < g(p)\}$, with $\bar{p}_0 = \sup \mathcal{C}_0$; $\mathcal{C}_1$ (with $\bar{p}_1 = \sup \mathcal{C}_1$) is the set $\{V^D < g\}$ of the model with $\kappa = 1$. The *observability-blind policy* NB^D applies to the true belief the posture and stopping rule of Proposition 5(ii) computed with $\kappa = 1$; $J^{\text{NB}}(p_0; \kappa)$ is its expected cost in the model with the true κ . Model D is a limit of Definitions 1–7 (period length to zero at fixed separation per unit

time, $T \rightarrow \infty$, ~~verify removed~~); that limit motivates the model and is not a statement of this paper.

Proposition 5 (Index law in the diffusion model). Under Definition 12 and Assumptions 1 and 4, with $d > 0$ and $r_K > 0$, let $c^*(p) = \min\{pr_K, (\delta + \rho pr_K)/\kappa^2\}$ ($c^*(p) = pr_K$ if $\kappa = 0$), and let $\tilde{P}$ be the belief of the monitoring model, $d\tilde{P}_u = d\tilde{P}_u(1 - \tilde{P}_u)dW_u$, $\tilde{P}_0 = p$, with W a standard Brownian motion. (i) (*Reduction to optimal stopping.*) $V^D(p) = \inf_{\sigma} \mathbb{E}_p \left[\int_0^{\sigma} c^*(\tilde{P}_u) du + g(\tilde{P}_{\sigma}) \right]$, the infimum over a.s. finite stopping times of $\tilde{P}$. (ii) (*Index.*) If $\delta > 0$, $\rho > 0$ or $\kappa = 0$, there is an optimal policy that stops when p_t leaves the continuation set of (i) and, before stopping, throttles at belief p if and only if $\kappa^2 > \rho + \delta/(pr_K)$, and monitors otherwise. When $\kappa^2 > \rho$, the throttled beliefs are those above $p_{\text{thr}} = \delta/((\kappa^2 - \rho)r_K)$. If $\delta = \rho = 0 < \kappa$, $V^D(p) = pC_K + (1 - p)H_F$ and, for $p \in (0, 1)$, no policy is optimal. (iii) (*Critical factor.*) $\mathcal{C}_0 = (\underline{p}_0, \bar{p}_0)$ is an open interval that contains q^* . For $p_0 \in \mathcal{C}_0$, $G^D(\kappa) > 0$ if and only if $\kappa^2 > \rho + \delta/(\bar{p}_0 r_K)$; hence $\kappa_D^* = \min \left\{ 1, \sqrt{\rho + \delta/(\bar{p}_0 r_K)} \right\}$, and $\kappa_D^* = \sqrt{\rho}$ when $\delta = 0$, whatever the loss table, d and $p_0 \in \mathcal{C}_0$. In particular, if $\kappa = r$ and $\rho = r^\alpha$ for some $r \in (0, 1)$ and $\alpha > 0$ (a scaling throttle, Proposition 4(i), with harm of order α in the injected flow), then $G^D(\kappa) > 0$ requires $\alpha > 2$. (iv) (*Observability-blind operator.*) Suppose $\rho < 1$, $\delta > 0$ or $\rho > 0$, $p_{\text{NB}} = \delta/((1 - \rho)r_K) < \bar{p}_1$ and $p_0 \in \mathcal{C}_1$. Then $\mathcal{C}_1 = (\underline{p}_1, \bar{p}_1)$ is an open interval containing q^* ; $J^{\text{NB}}(p_0; \kappa)$ is continuous and strictly decreasing in $\kappa \in (0, 1]$, tends to $+\infty$ as $\kappa \downarrow 0$, and equals $J(p_0)$ for the solution J of $\frac{1}{2}d^2p^2(1 - p)^2J''(p) = -\hat{c}_\kappa(p)$ on $\mathcal{C}_1$ with $J = g$ at $\underline{p}_1$ and $\bar{p}_1$, where $\hat{c}_\kappa(p) = pr_K$ for $p \leq p_{\text{NB}}$ and $(\delta + \rho pr_K)/\kappa^2$ for $p > p_{\text{NB}}$. Hence $\{\kappa \in (0, 1] : J^{\text{NB}}(p_0; \kappa) > V^{D, \text{KS}}(p_0)\} = (0, \kappa_{\text{NB}})$ for a unique $\kappa_{\text{NB}} \in (0, 1]$.

Proposition 5 is claim 7(a). Its proof has been checked by the author and not yet independently verified; its idea is given below, and the full proof is not included in this version.

In the diffusion model a throttled period yields information at the rate κ^2 relative to monitoring, since the Kullback–Leibler divergence between two Gaussian laws is quadratic in the distance between their means. Part (ii) says that, when the throttle has a disruption cost or leaves some harm ($\delta > 0$ or $\rho > 0$), it is worth using at a belief exactly when its running cost per unit of information, $(\delta + \rho pr_K)/\kappa^2$, is below monitoring’s, pr_K . The name “index law” is shorthand for this comparison: it is a cost per unit of information in a single undiscounted stopping problem, not a Gittins or bandit index, since

there is no discounting, no competing project and no retirement reward. The hypothesis excludes one degenerate case: a throttle with $\delta = \rho = 0 < \kappa$ is free and still informative, so from any prior belief in $(0, 1)$ throttling until the belief is close to 0 or 1 comes arbitrarily close to the cost $pC_K + (1-p)H_F$ of the chord of the terminal cost, and no policy attains it. Because the harm accrues only under a compromise, the comparison depends on the belief: the operator monitors at low beliefs and throttles at high ones. Part (iii) turns this into a closed form for the critical factor, κ_D^* , which does not depend on the loss table, on d or on the prior when there is no disruption cost, and it gives a sharp consequence for the scaling throttle of Proposition 4(i): since the throttle gives $\kappa = r$, information falls quadratically in the throttle factor, and such a throttle is worth using only if harm falls faster than quadratically in the throttled flow. Part (iv) makes the observability-blind operator of claim 4 exact in this model: an operator who plans as if throttling kept full discriminability pays a cost that grows without bound as κ falls to zero, so it does worse than the best kill-switch for every κ below a threshold $\kappa_{NB} > 0$.

The technique is not new. For a Brownian drift with a controlled signal intensity and a running cost that does not depend on the belief, Campbell et al. (2025) show that the optimal control is the constant minimiser of the cost per unit of information rate, which reduces the problem to optimal stopping; Harrison and Sunar (2015) study a binary unknown learnt through several modes, each with a cost rate and a signal quality, before a decision to invest or abandon. What Proposition 5 adds is a running cost that depends on the belief, because harm accrues only under a compromise, so that the optimal posture is a switch in the belief rather than a constant; an action that acts on harm and on information at once, through ρ and κ ; the closed form of part (iii); and the misspecified planner of part (iv).

Idea of the proof. The cumulative information $A_t = \int_0^t \kappa_{x_s}^2 ds$ is a time change under which the belief is the monitoring belief run at information time (Campbell et al., 2025). Each posture's running cost is at least κ_x^2 times the lower envelope c^* , so every policy costs at least the value of optimal stopping of the monitoring belief with running cost c^* (Peskir and Shiryaev, 2006); running real time at κ^{-2} per unit of information time with the posture of part (ii) attains this bound, and the free boundary of the stopping problem gives parts (iii) and (iv).

Model D is a limit of the discrete model (period length to zero at fixed separation per unit time, no horizon, no **verify**); this motivates it, and is not a result of the paper. The discrete model has a deadline, which makes

information worth less near the horizon, so its κ^* need not equal κ_D^* . Whether the law locates the discrete κ^* on the calibrated region is claim 7(b), tested separately; it does not (Section 5.6).

4.7. *What the results do not give*

Weak dominance of the graded policy over the best kill-switch, $\Gamma(p_0) \geq 0$, holds by construction, because KS^* is in the graded policy’s class; it is not a finding. The results above do not say, for the discrete model, when the gain is strict, where κ^* lies inside $(0, 1]$, or how much an observability-blind operator loses. They also say nothing about the arrangement of the middle region beyond Proposition 2. These are the questions of Section 5.

In the three-state model, Lemma 1, Proposition 1(ii), Theorem 1(i), (iii)–(iv) and Proposition 2(iv) hold on the 2-simplex. Verification reads authorization, which is valid under both a fault and a benign shock: under Definition 4 with $t_B = t_F$, its outcome has the same law under F and B . Only order flow separates these two causes, so where a compromise has been ruled out the throttle’s observability price is paid in the F – B separation alone. What this changes in the optimal policy is part of claim 5, tested numerically.

5. Numerical study

The results of Section 4 hold for every parameter value but leave open the paper’s substantive claims. This section describes an exact computation that tests them. The design, the tolerance, the materiality margin, the calibrated region and the criteria that would support, weaken or refute each claim were fixed before any result existed, in a written experiment brief (brief B001, dated 29 September 2026, with four amendments (29–30 September 2026), all written before any result of the sweep, none of which changes a prediction or a criterion; see the data and code availability statement). The amendments change the following. Amendment 1, an answer to the feasibility review before any computation, narrows two arms that lie outside the calibrated region (the direct sweep over the separation d and the sensitivity with $H_F = 0$) to one slice of the region, confirms how the verification delay is computed (the stage cost does not depend on ω ; which observations arrive during the delay; when verification is available) and sets the compute budget. Amendment 2, written after a one-cell pilot on a superseded data basis and using only its sanity checks, timing and convergence counts, fixes the quadrature rule (the Bayes-consistent rule of Proposition 1(iii) for the

two-state model), the grid-refinement schedules and the convergence rule for claim 5. Amendment 3 moves the inputs to the companion study’s final data basis and defines the calibrated cell set $\mathcal{F}$ by rule, as the cells without a supported improvement over the best constant policy, restating every criterion on $\mathcal{F}$ in place of a fixed list of cells. Amendment 4 is clarifications only: the belief used by the observability-blind policy in throttled periods (Definition 9(v)), grid convergence for every reported value (Definition 8), and notation. The brief is held under version control, whose commit history timestamps it and each amendment; it was not lodged with an external registry.

5.1. Questions

The paper makes seven claims. Claims 1–5 are the first five questions below, asked on a calibrated region of parameters and computed exactly by backward induction on the belief grid (Definitions 1–10); claims 3 and 4 are the two forms of the observability price (Section 6). Claim 2 is a corollary of Proposition 3, whose part (iv) already gives weak dominance of the graded policy over the one-shot rule; the computation reports the magnitude of the gap. Claim 6, on where $\kappa < 1$ comes from, is Proposition 4 (Section 3.7) and needs no computation. Claim 7 has two parts: part (a), the cost-per-information law, is Proposition 5 (Section 4.6), a result in the diffusion model; part (b), whether that law carries over to the discrete, finite-horizon model on the calibrated region, is the last question below. It is tested by a second brief (B002), which post-processes the output of the first, adds one slice without disruption cost, and fixed its own tolerances and bands before any result (Section 5.6).

1. Is the graded gain $\Gamma(p_0)$ strictly positive and material, and how much of it is the throttle’s share rather than verification’s? (Claim 1.)
2. When order flow is informative, how large is the gap between Paper 2’s one-shot rule P2 and the optimal graded policy, and is it material? (Claim 2, a corollary of Proposition 3.)
3. Is the critical discriminability factor κ^* below 1 on the calibrated region, so that the throttle has value there for a discriminability factor high enough? (Claim 3; $\kappa^* > 0$ at every point of the design is Proposition 6.)
4. For which κ does the observability-blind policy NB cost more than the best kill-switch KS*? (Claim 4.)

5. With a benign third cause, does modelling it change the optimal policy and its cost materially? (Claim 5.)
6. Does the law of Proposition 5 locate the κ^* of the discrete, finite-horizon model, and does the closed-form policy it suggests recover most of the graded gain? (Claim 7(b); second brief.)

5.2. Calibrated region

The region is built from Paper 1’s cost study (Mejrissi, 2026b) and a box, fixed in the brief, for what Paper 1 does not determine; Table 1 lists every value of the design.

- *Cells.* The set $\mathcal{F}$ of Paper 1’s cells without a supported improvement over the best constant policy: the cells of Paper 1’s cost table in which the cluster-bootstrap interval over seeds does not support an improvement of the classifier over the best constant policy. $\mathcal{F}$ is read from the table by this rule, is not restricted to low priors, and every cell it contains is a full member of the calibrated region. The cells in which the classifier’s point estimate alone does not beat the best constant policy form a subset of $\mathcal{F}$; they are marked in every per-cell report, as a description, and no criterion is evaluated on them separately. Each cell supplies its loss table (C_K, C_F, H_K, H_F) , with the measured cost of leaving a fault, and its prior p_0 . The remaining cells of Paper 1’s table are computed as a contrast set, reported beside the calibrated region and not part of it.
- *Separation.* The equivalent-Gaussian calibration (Remark 1) fixes the per-period separation under monitoring, $d = d_W / \sqrt{n_W}$; throttling multiplies it by κ , which is swept from 0 to 1 in steps of 0.05 (21 values). The window separation d_W is backed out of the cell’s classification accuracy, and the number of periods n_W the window spans is a modelling choice, taken at $n_W \in \{1, 4, 16\}$. Because this calibration is a heuristic, d is also swept directly, beside it and outside the calibrated region.
- *Fixed box.* The horizon $T = 20$; the harm rate $r_K = \eta a/T$ with $\eta \in \{0.25, 1, 4\}$, so that total harm over the horizon is measured against what killing saves; the harm factor $\rho \in \{0, 0.25, 0.5\}$; the disruption cost $\delta = \zeta b/T$ with $\zeta \in \{0.1, 1\}$; the verification sensitivity and specificity $s = t_F \in \{0.9, 0.99\}$, its cost $c_v = 0.05 \min(a, b)$, and its delay $\ell_v = 1$, with no order-flow observation during the delay ($\omega = 0$, the

Table 1: Design of the numerical study, fixed in brief B001 before any result: the box of parameters not determined by Paper 1, the sweep of κ , and the constants of the decision criteria. Values given as lists are crossed with one another at every cell of $\mathcal{F}$.

Quantity	Value
Horizon T	20
Periods per window n_W ($d = d_W/\sqrt{n_W}$)	1, 4, 16
Harm rate $r_K = \eta a/T$, multiple η	0.25, 1, 4
Harm factor under throttling ρ	0, 0.25, 0.5
Disruption cost $\delta = \zeta b/T$, multiple ζ	0.1, 1
Discriminability factor κ	0 to 1 in steps of 0.05 (21 values)
Verification sensitivity and specificity $s = t_F$	0.9, 0.99
Verification cost c_v , as a fraction of $\min(a, b)$	0.05
Verification delay ℓ_v	1
Order flow observed during the delay ω	0
Grid tolerance $\epsilon = c\Xi(p_0)$, constant c	10^{-3}
Materiality share m	0.05
Resolution of κ^* (bisection)	0.01
Three-state against two-state κ^* , margin	0.05

analogue of Paper 2’s no-further-evidence assumption). With the values of n_W and κ above, each cell of $\mathcal{F}$ contributes 2268 points to the calibrated region.

- *Sensitivity, outside the calibrated region.* Other horizons, a longer verification delay, $\omega = 1$, asymmetric $s \neq t_F$, other verification costs, and a zero cost of leaving a fault.
- *Three-state set.* The cells of $\mathcal{F}$ with $C_B = C_F$ and $H_B = H_F$, and a scalar order-flow statistic in which the benign shock’s mean is placed relative to the fault’s so that the F – B separation is a swept fraction of d ; the share of the non-compromise prior mass on B is also swept.

5.3. Metrics and decision criteria

Each point is measured against two reference costs: the *oracle cost* $O(p_0) = p_0 C_K + (1 - p_0) H_F$ of an operator who knows the cause, and the *residual ambiguity* $\Xi(p_0) = V_0^{\text{KS}}(p_0, 0) - O(p_0)$, what the best kill-switch still loses to perfect information.

- *Grid convergence.* A point is grid-converged (Definition 8) if doubling N and n_q changes every reported value (the optimal value, the values of the restricted recursions and the expected cost of every evaluated policy) by at most the tolerance $\epsilon = 10^{-3} \Xi(p_0)$. A difference of two converged values within twice that tolerance is zero up to the grid tolerance.
- *Materiality.* The graded gain is material at a point if it recovers at least a share $m = 0.05$ of the residual ambiguity, $\Gamma \geq m \Xi(p_0)$; the gap to P2 is material if it recovers the same share of what P2 loses to the oracle; in the three-state model, a difference is material if it recovers the same share of the three-state residual ambiguity of KS^* .
- *Shares.* The throttle share $V_0^{\text{KS}} - V_0^{\text{KST}}$, the verify share $V_0^{\text{KS}} - V_0^{\text{KSV}}$ and their interaction are reported beside Γ .
- *Critical factor.* At each point, κ^* is classed as 0, 1 or interior from G at the ends of the sweep, and located by bisection to a resolution of 0.01 when interior. A decrease of G in κ beyond the tolerance would contradict Theorem 1 and is treated as an error of the solver, not as evidence. The disruption cost is positive at every point of the design, so by Proposition 6 the class $\kappa^* = 0$ is empty; a point in it is likewise treated as an error of the solver.
- *Blind operator.* κ_{NB} is the largest κ of the sweep at which J^{NB} exceeds V_0^{KS} beyond the tolerance.
- *Beside cost.* False-kill and missed-containment rates, expected time to a terminal action and expected throttled time, for every policy; optimal-action maps over belief and time for a representative point of each cell, and on the simplex for the three-state set.

Each claim is supported, weakened, refuted or found inconclusive by criteria fixed in advance. Claim 1 is refuted if the graded gain is zero up to the tolerance at every point, and weakened if it is material at no more than half of the points, if it comes entirely from verification, or if its material points lie only outside the calibrated region. Claim 2 is refuted if the gap between P2 and the optimal graded policy is zero up to the tolerance at every point with $d > 0$, and weakened if it is material at no more than half of those points. Claim 3 is refuted if $\kappa^* = 1$ at every point of the calibrated region,

and weakened if $\kappa^* < 1$ only outside it. The outcome $\kappa^* = 0$ at every point cannot occur, since the disruption cost is positive (Proposition 6). Claim 4 is refuted if NB never costs more than KS^* beyond the tolerance, and weakened if it does so only outside the calibrated region. Claim 5 is weakened, and the three-state extension reported as a method rather than a result, if the value of modelling the benign cause (the cost, in the three-state model, of the policy that is optimal when the benign shock is treated as a fault, minus the three-state optimal value) is material at no point and the three-state and two-state κ^* agree within 0.05 at every point; it has no refutation criterion, because exact solvability on the simplex is a property of the method, checked by grid convergence and by the check of Section 5.4. A claim is *inconclusive* if more than a quarter of its points are not grid-converged after two doublings or straddle the materiality threshold. Negative outcomes are reported beside positive ones.

5.4. Checks before any result

Three results of Section 4 are checked on the solver’s output before the sweep is read: concavity of the two-state grid values under the Bayes-consistent rule (Proposition 1(iii)); monotonicity of G in κ (Theorem 1); and, at $d = 0$, equality of the optimal value with the cost of Paper 2’s rule, with S computed on the grid from Paper 2’s definition directly (Proposition 3(ii), which holds for every horizon $T \geq \ell_v$; the check is run at $T = \ell_v$, its special case in which the horizon equals the verification delay). In the three-state model, the value of verification on the F – B edge must be zero up to the tolerance. A failure of any of these is an error of the implementation.

5.5. Results

Every figure below is read from the generated tables of the sweep; each outcome is the one given by the criteria of Section 5.3, and negative outcomes are reported beside positive ones. Table 2 gives the outcome of each claim, Table 3 the figures per cell of $\mathcal{F}$. The sweep used 10.4 CPU-hours in 2 runs.

Checks and convergence. No check of Section 5.4 failed. Concavity was violated at 0 of 44100 grid values; G decreased in κ beyond the tolerance at 0 two-state and 0 three-state points; the optimal value matched the cost of Paper 2’s rule at $d = 0$ (check passed: yes); the value of verification on the F – B edge exceeded the tolerance at 0 points; and 0 κ -groups were classed $\kappa^* = 0$. Of the 22680 points of the calibrated region, 20256 are grid-converged after

Table 2: Outcome of claims 1–5 on the calibrated region, by the criteria fixed in brief B001 before any result. Row: the brief’s label of the claim.

Row	Claim	Outcome	Deciding figures
1	1	supports	material at 16125 of 20256 converged points (0.796); lowest cell 0.556
1b	2	weakens	material at 3707 of 20256 converged points (0.183)
2a	3	supports	κ^* interior at 10, = 1 at 910, = 0 at 0 of 920 converged groups
2b	4	supports	κ_{NB} defined in 8 groups
4	5	inconclusive	227 of 600 points unusable, 16 usable points straddling (share 0.405)

Table 3: Calibrated region, one line per cell of $\mathcal{F}$. Points: points of the cell; Conv.: grid-converged; Strad.: straddling the materiality line; share: material over converged; the last four columns count κ -groups (a point without its value of κ).

Cell of $\mathcal{F}$	Points	Conv.	Strad.	Γ material (share)	Gap P2 material	κ -groups conv.	$\kappa^* = 1$	Interior	κ_{NB} defined
Original design (V1), contrast, $p_0 = 0.05$	2268	1728	2	1050 (0.608)	606	73/108	73	0	0
Original design (V1), contrast, $p_0 = 0.1$	2268	1843	1	1565 (0.849)	699	79/108	74	5	4
Study 5 (V2), contrast, $p_0 = 0.05$	2268	1859	0	1144 (0.615)	442	83/108	83	0	0
Study 5 (V2), contrast, $p_0 = 0.1$	2268	1939	103	1535 (0.792)	776	82/108	79	3	3
Study 5 (V2), summary-matched, $p_0 = 0.05$	2268	2268	0	1260 (0.556)	252	108/108	108	0	0
Study 5 (V2), summary-matched, $p_0 = 0.1$	2268	2242	252	1621 (0.723)	130	105/108	105	0	0
Study 5 (V2), summary-matched, $p_0 = 0.2$	2268	2268	0	2268 (1.000)	0	108/108	108	0	0
Study 5 (V2), summary-matched, $p_0 = 0.3$	2268	2268	0	2268 (1.000)	0	108/108	108	0	0
Study 5 (V2), summary-matched, $p_0 = 0.5$	2268	2268	0	2268 (1.000)	3	108/108	107	1	0
Fourth design (V4), contrast, $p_0 = 0.05$	2268	1573	0	1146 (0.729)	799	66/108	65	1	1

two doublings and 2424 are not; with the 358 converged points that straddle the materiality line, they are a share 0.123 of the points, below the quarter that would make claim 1 inconclusive. Unconverged points are concentrated in the contrast cells, up to a share 0.306 of one cell.

Claim 1: supported. The graded gain is material at 16125 of the 20256 converged points (share 0.796), and at more than half of the converged points in each of the 10 cells of $\mathcal{F}$; the lowest cell share is 0.556 (Study 5 (V2), summary-matched, $p_0 = 0.05$). The gain does not come entirely from verification, but the throttle carries little of it: the throttle share exceeds the grid tolerance at 644 of the 16125 material points. The gain is therefore not a throttling result. In the contrast set and the direct sweep of d , reported beside the region, the gain is material at 3959 points. The operating characteristics of each policy (false-kill and missed-containment rates, time to a terminal action, throttled time) and the optimal-action maps are not

tabulated in this version.

Claim 2: weakened. The gap between Paper 2’s one-shot rule and the optimal graded policy is positive beyond the tolerance at 4681 points, but material at only 3707 of the 20256 converged points (share 0.183), no more than half, which weakens the claim by its own criterion. Per cell the material share ranges from 0.000 (Study 5 (V2), summary-matched, $p_0 = 0.2$) to 0.508 (Fourth design (V4), contrast, $p_0 = 0.05$); in 2 cells it is material nowhere. The value of sequential evidence over the one-shot rule is thus real but, on most of the region, below the materiality margin.

Claim 3: supported, on few groups. Of the 1080 κ -groups of the region, 920 are converged (160 are not). Among them $\kappa^* = 1$ in 910 and $\kappa^* < 1$ in 10, located between 0.219 and 0.769; these few groups come from few cells (Table 3). The claim is supported, since $\kappa^* < 1$ on a non-empty part of the calibrated region, but in every other converged group the throttle has no value at any factor below one. The brief also predicted that κ^* increases with δ and ρ . On the converged pairs κ^* is non-decreasing in δ (422 of 422 pairs) and in ρ (596 of 596), but strictly increasing in only 7 and 4 of them, most ties being at $\kappa^* = 1$: the dependence is shown to be non-decreasing, not increasing.

Claim 4: supported, on the same groups. The observability-blind policy costs more than the best kill-switch beyond the tolerance, for κ below a κ_{NB} , in 8 κ -groups of the region, 8 of them among the 10 with $\kappa^* < 1$; in the contrast set and the direct sweep it does so in 0. The region where ignoring the observability price is costly is non-empty on the calibrated region, as claimed, and as narrow as the region where the throttle has value at all.

Claim 5: inconclusive at this power. Of the 600 three-state points, 227 are unusable (205 three-state values not converged, and 22 whose matched two-state value is not converged), and 16 usable points straddle the materiality line: a share 0.405 above the quarter fixed in advance (Table 4). Claim 5 is therefore inconclusive at this power, and nothing is read from the usable points. The check of verification on the F – B edge, reported above, is a check of the solver, not evidence for the claim.

5.6. Does the diffusion law carry over? (Claim 7(b))

The second brief post-processes the converged output of the sweep; it was written, with its tolerances and bands, before any result of the sweep

Table 4: Three-state set, one line per cell. Usable: the three-state value and the matched two-state value are both grid-converged; Straddling: over all points.

Cell of $\mathcal{F}$	Points	Converged	Usable	Straddling
Original design (V1), contrast, $p_0 = 0.05$	60	1	1	0
Original design (V1), contrast, $p_0 = 0.1$	60	17	13	10
Study 5 (V2), contrast, $p_0 = 0.05$	60	54	48	10
Study 5 (V2), contrast, $p_0 = 0.1$	60	23	11	10
Study 5 (V2), summary-matched, $p_0 = 0.05$	60	60	60	0
Study 5 (V2), summary-matched, $p_0 = 0.1$	60	60	60	10
Study 5 (V2), summary-matched, $p_0 = 0.2$	60	60	60	0
Study 5 (V2), summary-matched, $p_0 = 0.3$	60	60	60	0
Study 5 (V2), summary-matched, $p_0 = 0.5$	60	60	60	0
Fourth design (V4), contrast, $p_0 = 0.05$	60	0	0	25

existed. At each point it reads the upper edge $\bar{p}_0$ of the throttle-free continuation set and forms the law’s prediction $\hat{\kappa}^* = \min\{1, \sqrt{\rho + \delta/(\bar{p}_0 r_K)}\}$ (Proposition 5(iii)). It has four parts. (a) *Location*: on the κ -groups where the law predicts that throttling has value ($\hat{\kappa}^* < 1$), the share at which κ^* lies within the brief’s location tolerance of $\hat{\kappa}^*$; it supports claim 7(b) above three quarters, weakens it between a quarter and three quarters, and refutes it at a quarter or less. (b1) *Scaling diagonal*: whether $G(\rho)$ is zero up to the grid tolerance, as the law predicts for a scaling throttle with linear harm. (b2) *No disruption cost*: on a new slice with $\delta = 0$, whether κ^* lies within the same tolerance of $\sqrt{\rho}$. (c) *Closed-form policy*: the policy IDX takes the throttle-free optimal action where it is terminal or **verify** and otherwise throttles if and only if $\kappa^2 > \rho + \delta/(p r_K)$; at the points where the graded gain is material, the share at which its regret is at most a fifth of the gain. (d) *Band order*: at the points where the optimal policy throttles at some reachable state, the share at which no monitored belief lies above a throttled one. By claim 7, (b) is supported if (a) and (c) fall in their support bands, weakened if either falls in its weaken band, and refuted if (a) falls in its refute band. Table 5 gives the outcome of each part.

Claim 7(b): refuted. The law does not locate the discrete κ^* . Among the 690 converged κ -groups with $\hat{\kappa}^* < 1$, κ^* lies within the tolerance at 39 (share 0.057; per cell from 0.037 to 0.131), in the refute band fixed before the result; 160 unconverged groups are excluded. By the rule stated in claim 7,

Table 5: Outcome of each part of the test of claim 7(b), by the criteria fixed in brief B002 before any result. n : usable κ -groups (parts a, b2) or points (b1, c, d); Share: the share the criterion is read on. For part d, “supports” means that the band order holds.

Part	Outcome	n	Share
a	refutes	690	0.057
b1	supports	636	0.000
b2	weakens	18	0.556
c	supports	5905	0.994
d	supports	920	0.880

claim 7(b) is refuted. The miss is one-sided. Where the law predicts that throttling has value, the discrete model almost never uses the throttle: $\kappa^* < 1$ in only 10 of the 920 converged groups (claim 3). Where the law predicts that it has none ($\hat{\kappa}^* = 1$, 230 groups), the discrete model throttles in 0: the law never misses a throttle that the discrete model uses.

The other parts. They are reported beside the refutation, with the same weight. (b1) *Supports*: $G(\rho)$ exceeds twice the tolerance at 0 of 636 points, so a scaling throttle with linear harm has no value, as the law predicts. (b2) *Weakens*: with no disruption cost, κ^* lies within the tolerance of $\sqrt{\rho}$ in 10 of 18 groups (share 0.556). (c) *Supports*: IDX’s regret is at most a fifth of the graded gain at 5871 of the 5905 points where the gain is material (share 0.994; lowest cell 0.978), and IDX is worse than never throttling at 0 points. This is not evidence that the law locates κ^* : the throttle carries little of the graded gain on the region (claim 1), so a policy can recover most of the gain whatever it does with the throttle, and how often IDX throttles has not been measured. (d) *Holds*: the band order holds at 810 of the 920 points where the optimal policy throttles (share 0.880; lowest cell 0.833), so where the discrete model throttles, it throttles at high beliefs and monitors at low ones, as in the diffusion model. Claim 7(a), Proposition 5, is a result about the diffusion model and is unaffected.

6. Discussion

What is claimed, and at what strength. The structural results of Section 4 hold for every parameter value; of them, Lemma 1, Propositions 1–2 and Theorem 1 are classical results specialised to this model, Proposition 6 is specific

to graded containment, and Proposition 5 holds in a diffusion model. The paper’s numerical claims do not hold for every parameter value: strict dominance of the graded policy over the best kill-switch, a critical discriminability factor below one, the loss of an observability-blind operator, the effect of a benign third cause and the carry-over of Proposition 5 to the discrete model are claims about a calibrated region, and the gain over the one-shot rule is a corollary of Proposition 3 whose size is measured there. The computation, fixed in advance, of Section 5 supports strict dominance, $\kappa^* < 1$ and the loss of the blind operator, weakens the gain over the one-shot rule, is inconclusive on the benign cause, and refutes the carry-over of Proposition 5 as a way to locate κ^* . Weak dominance is not a finding, since the kill-switch is one of the graded policies. The observability price is stated in two forms that do not collapse into each other. For an optimal operator it is a threshold: below κ^* the throttle has no value (Theorem 1), which says when to throttle, not that throttling is harmful; the threshold is strictly positive when throttling has a disruption cost (Proposition 6), and in the diffusion model it is where the throttle’s cost per unit of information equals monitoring’s (Proposition 5). For an operator who plans as if throttling kept full discriminability, it is a loss: in the diffusion model this operator’s cost grows without bound as κ falls (Proposition 5(iv)); in the discrete model the policy NB does worse than the best kill-switch on part of the calibrated region, so ignoring the price is costly there, not merely suboptimal.

What the computation shows, and how far. On the calibrated region the graded gain is large and frequent, but the throttle carries little of it, and $\kappa^* < 1$ and the blind operator’s loss are found on few κ -groups from few cells (Section 5.5). The observability price is therefore shown to exist on the region, not to be common there. The refutation of claim 7(b) is one-sided: where the diffusion law predicts that throttling has value, the discrete model almost never throttles, and it never throttles where the law predicts no value. One possible reason, a hypothesis that this paper does not test, is that verification and the finite horizon, both absent from the diffusion model, supply the information that the throttle would buy.

The discriminability factor. The POMDP treats ρ and κ as independent parameters and κ as a constant. Proposition 4 shows that neither simplification is innocent in Paper 1’s linear model: a scaling throttle ties κ to the scaling factor, and a feedback-suppressing throttle started after the onset has

a transient in which the residual feedback keeps the throttled observations informative. The constant κ of the POMDP is exact for a throttle in force from the onset and is otherwise an idealisation that holds once the transient has decayed. Throttles that make the causes easier to tell apart, for instance because a persistent adversary pushes against the limit, are excluded by Assumption 4; they would turn **throttle** into an information-gathering action.

Calibration. The separation is backed out of a classifier accuracy through a Gaussian benchmark that Paper 1’s simulated constructions were never shown to satisfy, and it is a separation for a whole window, not for a period. It is therefore labelled a heuristic, converted to a per-period value under a stated window length, and accompanied by a direct sweep of d . Assumption 2 also ignores the growth of separation with time that state feedback produces in Paper 1’s model.

Known model, non-strategic cause. The operator knows every parameter and the cause does not react (Assumption 5). The contribution is the structure of the trade-off between harm, disruption and evidence, which needs exact solutions; robustness to a misjudged κ is what the blind policy NB measures. Learning the model online, and an adversary who adapts to the operator’s actions, are outside the paper’s scope; the strategic case is the subject of Paper 4 (Mejrissi, 2026a).

More than two candidate causes. The three-state extension is the one place where verification is structurally partial: it separates a compromise from the other two causes but not a fault from a benign shock. It is solved exactly on the simplex, with the structural results that carry over (Section 4.7). Whether it changes the policy materially, or only repeats the two-state picture on a larger grid, is claim 5, which the computation leaves inconclusive at this power: too many three-state points are not grid-converged or straddle the materiality line. Until a finer computation decides it, the extension is a method, not a result.

7. Conclusion

Responding to an ambiguous incident on a trading gateway is a sequential decision in which containment has a price in evidence. We have modelled it as a POMDP over the hidden cause, with a graded ladder of actions and a throttle that reduces harm and discriminability together, and solved it

exactly on a belief grid. The value function is concave, the kill and leave regions are intervals, and the optimal value is monotone in the discriminability factor, so that there is a single critical factor κ^* below which an optimal operator need never throttle. These results are classical, specialised to the model. Two are specific to graded containment: a throttle that blinds too much is never used, so $\kappa^* > 0$ when throttling has a disruption cost; and, in a diffusion model where the throttle has a disruption cost or leaves some harm, it is worth using exactly where its cost per unit of information is below monitoring's, which gives κ^* in closed form, rules out scaling throttles unless harm falls faster than quadratically in the flow, and makes the loss of an operator blind to the observability price unbounded as κ falls. Paper 2's one-shot verification rule is the special case with uninformative order flow. In Paper 1's linear-Gaussian model, a throttle that suppresses the incident's feedback from the onset leaves no discriminability for pairs confounded at the output level, and a scaling throttle keeps exactly its scaling factor. An exact computation with criteria fixed in advance shows that, on a region calibrated from Paper 1's cells, the graded policy strictly and materially beats the best kill-switch at most points, with little of the gain from the throttle; that $\kappa^* < 1$, and an operator who ignores the observability price does worse than the kill-switch, on a small part of the region; that the gain over the one-shot rule is material at no more than half of the points; and that the effect of a benign third cause is inconclusive at this power. The diffusion law does not locate the discrete κ^* : that claim is refuted by the test fixed before the result. Of the law's other predictions, those for scaling throttles and for the band order of the optimal policy hold, and $\kappa^* = \sqrt{\rho}$ without disruption cost is weakened.

Data and code availability

The solver that computes the optimal policies and the reference policies of Section 5, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author's project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). The experiment brief that fixes the design and the decision criteria of Section 5 (brief B001, file `papers/paper3/research/briefs/B001_graded-vs-killswitch-solver.md`) and its amendments are held in the same repository, whose commit history timestamps each of them, and will

be deposited with it. Until then they are available from the corresponding author on reasonable request. The loss tables, priors and classification accuracies that calibrate the numerical study are those of the companion study (Mejrissi, 2026b), whose deposit they share.

CRediT authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Appendix A. Proofs

This appendix proves the results of Section 4, in the order in which they are stated there (Proposition 5 excepted, see below), and then Proposition 4 of Section 3.7, whose part (iv) gives explicit instances with $\kappa_W < 1$, $\kappa_W = 1$ and $\kappa_W > 1$.

Status of the proofs. The results differ in how far their proofs have been checked, and we state this for each.

- *Independently verified:* Lemma 1, Propositions 1–4 and Theorem 1. Each statement was proved a second time, independently, from the statement, its definitions and its assumptions alone, without access to the proofs below or to the rest of the paper; each of these proofs was complete, with no gap. For Proposition 1 the verification covers the recursion of Definition 7, with the full action set.
- *Checked by the author only:* Propositions 5 and 6. The full proof of Proposition 5 is not included in this version; Section 4.6 gives its idea. For Proposition 6, part (iii) uses the continuity of the value V^{KSV} of a restricted recursion of Definition 9, which Proposition 1, as stated and as verified, does not cover; the proof ends with this dependency and a proposed closure (by part (ii), $V^{\text{KSV}} = V(\cdot; 0)$, to which Proposition 1 applies). The closure has not been independently verified.

Proof of Lemma 1. The perspective function. On the closed quadrant $\mathcal{Q} = [0, \infty)^2$ put $F(u, w) = (u + w) f(u/(u + w))$ if $u + w > 0$ and $F(0, 0) = 0$. Then $|F(u, w)| \leq (u + w) \|f\|_\infty$, F is positively homogeneous ($F(cz) = cF(z)$ for $c \geq 0$), and, for every o and $p \in [0, 1]$,

$$\Lambda_p(o) f(q_p(o)) = F(p \lambda_K(o), (1 - p) \lambda_F(o)),$$

including where $\Lambda_p(o) = 0$ (both arguments are then zero). So $(T_\lambda f)(p) = \int_{\mathcal{O}} F(p \lambda_K(o), (1 - p) \lambda_F(o)) \nu(do)$, and the integrand is bounded by $(\lambda_K + \lambda_F)(o) \|f\|_\infty$, which is ν -integrable.

(i) F is concave on $\mathcal{Q}$ when f is. Let $z_i = (u_i, w_i) \in \mathcal{Q}$, $s_i = u_i + w_i$, $\lambda \in (0, 1)$. If $s_1 = 0$ then $z_1 = 0$ and $F(\lambda z_1 + (1 - \lambda) z_2) = (1 - \lambda) F(z_2) = \lambda F(z_1) + (1 - \lambda) F(z_2)$ by homogeneity; similarly if $s_2 = 0$. Otherwise let $s = \lambda s_1 + (1 - \lambda) s_2 > 0$ and $\mu = \lambda s_1 / s \in (0, 1)$. Then

$$\frac{\lambda u_1 + (1 - \lambda) u_2}{s} = \mu \frac{u_1}{s_1} + (1 - \mu) \frac{u_2}{s_2},$$

and concavity of f gives

$$F(\lambda z_1 + (1 - \lambda) z_2) = s f\left(\mu \frac{u_1}{s_1} + (1 - \mu) \frac{u_2}{s_2}\right) \geq s \mu f\left(\frac{u_1}{s_1}\right) + s(1 - \mu) f\left(\frac{u_2}{s_2}\right) = \lambda F(z_1) + (1 - \lambda) F(z_2).$$

For fixed o , $p \mapsto (p\lambda_K(o), (1-p)\lambda_F(o))$ is affine into $\mathcal{Q}$, so the integrand is concave in p ; integrating the pointwise concavity inequality over o (all terms integrable) shows that $T_\lambda f$ is concave.

(ii) If f is continuous on $[0, 1]$, F is continuous on $\mathcal{Q} \setminus \{0\}$ as a composition of continuous maps, and at 0 because $|F(u, w)| \leq (u + w)\|f\|_\infty$. Hence, for each o , the integrand is continuous in p ; it is dominated by the integrable $(\lambda_K + \lambda_F)\|f\|_\infty$, so $T_\lambda f$ is continuous by dominated convergence.

(iii) Fix p and write $u'(o') = p\lambda'_K(o')$, $w'(o') = (1-p)\lambda'_F(o')$, $u(o) = p\lambda_K(o) = \int \mathcal{G}(o \mid o') u'(o') \nu'(do')$, $w(o) = (1-p)\lambda_F(o) = \int \mathcal{G}(o \mid o') w'(o') \nu'(do')$, and $c(o) = u(o) + w(o)$. We show, for every o ,

$$F(u(o), w(o)) \geq \int_{O'} \mathcal{G}(o \mid o') F(u'(o'), w'(o')) \nu'(do'). \quad (\dagger)$$

If $c(o) = 0$, both sides are zero: the left by definition, the right because $\mathcal{G}(o \mid \cdot)(u' + w') = 0$ ν' -a.e. and $|F(u', w')| \leq (u' + w')\|f\|_\infty$. If $c(o) > 0$, let $\mu_o(do') = \mathcal{G}(o \mid o')(u' + w')(o') \nu'(do')/c(o)$, a probability measure, and $x(o') = u'(o')/(u' + w')(o') \in [0, 1]$ (defined μ_o -a.e.). Then $u(o)/c(o) = \int x d\mu_o =: \bar{x}$, and by homogeneity and Jensen's inequality for the concave continuous f on $[0, 1]$,

$$F(u(o), w(o)) = c(o) f(\bar{x}) \geq c(o) \int f(x) d\mu_o = \int \mathcal{G}(o \mid o')(u' + w')(o') f(x(o')) \nu'(do'),$$

which is the right side of $(\dagger)$. Integrating $(\dagger)$ over o and exchanging the order of integration (Tonelli applies to $\mathcal{G}(u' + w')\|f\|_\infty$, whose double integral is $\|f\|_\infty \int (u' + w') d\nu' < \infty$), and using $\int_O \mathcal{G}(o \mid o') \nu(do) = 1$,

$$(T_\lambda f)(p) \geq \int_{O'} F(u'(o'), w'(o')) \nu'(do') = (T_{\lambda'} f)(p).$$

(iv) If $Y' \sim \mathcal{N}(m, I_k)$ and $\varepsilon \sim \mathcal{N}(0, I_k)$ is independent of Y' , then $Y = rY' + \sigma_r \varepsilon \sim \mathcal{N}(rm, (r^2 + \sigma_r^2)I_k) = \mathcal{N}(rm, I_k)$, and the conditional density of Y given $Y' = y'$ is $\mathcal{G}(\cdot \mid y')$, which does not depend on m . Hence $\int \mathcal{G}(y \mid y') \phi_k(y' - m) dy' = \phi_k(y - rm)$ for every m .

Three-state model. Replace $\mathcal{Q}$ by $[0, \infty)^3$ and put $F(\alpha) = (\sum_M \alpha_M) f(\alpha / \sum_M \alpha_M)$, $F(0) = 0$. The proof of (i) is unchanged (the convex combination of two points of $\mathcal{S}_2$ with weights $\mu, 1 - \mu$ replaces that of two points of $[0, 1]$), and so is (ii). In (iii), $x(o') \in \mathcal{S}_2$, and Jensen's inequality $f(\int x d\mu_o) \geq \int f(x) d\mu_o$ holds for a concave continuous f on the compact convex set $\mathcal{S}_2$: such an f is the pointwise infimum of the affine functions $g \geq f$ (it is closed, proper and concave), and $\int f d\mu_o \leq \int g d\mu_o = g(\int x d\mu_o)$ for each of them. $\square$

Proof of Proposition 1. Every expectation in Definition 7 is a Bayes operator. For a posture x , the observation y has densities $\lambda_K^x = \phi(\cdot - \frac{1}{2}d_x)$, $\lambda_F^x = \phi(\cdot + \frac{1}{2}d_x)$ (Definition 3), the belief update of Definition 5 is Bayes' rule for them, and the expectation in $Q_t(\cdot, v, x)$ is over the predictive mixture Λ_p . Hence $\mathbb{E}_y[V_{t+1}(p', v)] = (T_{\lambda^x} V_{t+1}(\cdot, v))(p)$. For **verify**, the observation is $o = (z, y_1, \dots, y_{\ell_v})$ if $\omega = 1$ (only z if $\omega = 0$), with densities, with respect to counting measure on $\{+, -\}$ times Lebesgue measure,

$$\lambda_M^v(o) = P(Z = z \mid M) \prod_{i=1}^{\ell_v} \lambda_M^{\text{monitor}}(y_i),$$

by Assumption 2 (conditionally i.i.d. order flow) and Assumption 3 (Z conditionally independent of the order flow). Bayes' rule for λ^v gives the belief p'' of Definition 7 (applying the updates of Definition 5 one at a time gives the same posterior), so $\mathbb{E}[V_{t+\ell_v}(p'', 1)] = (T_{\lambda^v} V_{t+\ell_v}(\cdot, 1))(p)$. The same holds in the three-state model with the posteriors of Definition 5.

Boundedness. Every V_t is bounded: $V_T = R$ is, and if V_s is bounded for $s > t$, each Q_t is a bounded stage cost plus an operator T_λ applied to a bounded function, which is bounded by $\|V_s\|_\infty$ since $\int \Lambda_p d\nu = 1$.

(i) and (ii). Backward induction on t , for both values of v at once. At $t = T$, $V_T(\cdot, v) = R$ is the minimum of two affine functions, hence concave and continuous; in the three-state model $V_T(\pi, v) = \min\{\sum_M \pi(M)C_M, \sum_M \pi(M)H_M\}$, likewise. Let $t < T$ and assume the claim for all $s > t$. Then:

- $Q_t(\cdot, v, \text{kill})$ and $Q_t(\cdot, v, \text{leave})$ are affine;
- $Q_t(\cdot, v, \text{monitor}) = pr_K + T_{\lambda^{\text{monitor}}} V_{t+1}(\cdot, v)$ and $Q_t(\cdot, v, \text{throttle}) = \delta + \rho pr_K + T_{\lambda^{\text{throttle}}} V_{t+1}(\cdot, v)$ are an affine function plus a concave continuous function, by Lemma 1(i)–(ii) and the induction hypothesis;
- $Q_t(\cdot, 0, \text{verify}) = c_v + \ell_v pr_K + T_{\lambda^v} V_{t+\ell_v}(\cdot, 1)$ (when available, $t + \ell_v \leq T$) likewise.

$V_t(\cdot, v)$ is the minimum of finitely many concave continuous functions (the available set $\mathcal{A}_t(v)$ does not depend on the belief), hence concave and continuous. The three-state case is identical, with the stage costs $\sum_M \pi(M)r_M = \pi(K)r_K$ and Lemma 1 in its three-state form.

(iii) Under a Bayes-consistent rule, each expectation is T_λ for the finite measure $\nu = \sum_j \omega_j \delta_{o_j}$ and the likelihoods λ_M evaluated at the nodes;

Lemma 1(i) needs only $\lambda_K, \lambda_F \geq 0$ and ν -integrability, which hold. Write $I[g]$ for the piecewise-linear interpolant on G_N of a sequence g . A sequence on the uniform grid is concave ($g_{i-1} + g_{i+1} \leq 2g_i$) if and only if $I[g]$ is concave on $[0, 1]$, and the restriction to G_N of a concave function on $[0, 1]$ is a concave sequence. Backward induction: $\tilde{V}_T(\cdot, v)$ is the restriction of the concave R . If $\tilde{V}_s(\cdot, v')$ is a concave sequence for all $s > t$ and v' , then each $I[\tilde{V}_s(\cdot, v')]$ is concave, so each grid Q -value at $p \in G_N$ is the value at p of a concave function of $p \in [0, 1]$ (an affine stage or terminal cost plus T_λ of a concave function, Lemma 1(i)). Their restrictions to G_N are concave sequences, and the pointwise minimum of concave sequences is a concave sequence (if g, h are concave and $m = \min\{g, h\}$ with, say, $m_i = g_i$, then $m_{i-1} + m_{i+1} \leq g_{i-1} + g_{i+1} \leq 2g_i = 2m_i$). So $\tilde{V}_t(\cdot, v)$ is a concave sequence and $I[\tilde{V}_t(\cdot, v)]$ is concave.

Why (iii) needs its hypothesis. The two observations below are not part of the statement. (a) The per-component Gauss–Hermite rule of Definition 8 computes $p \sum_j w_j g(q_p(y_j^K)) + (1-p) \sum_j w_j g(q_p(y_j^F))$ with nodes $y_j^K = \frac{1}{2}d_x + x_j$, $y_j^F = -\frac{1}{2}d_x + x_j$. At a node y_j^K the rule carries the weight $p w_j$ but not the matching term $(1-p) w_j \lambda_F(y_j^K)/\lambda_K(y_j^K)$, so it is not of the form $\sum_j \omega_j \Lambda_p(o_j) g(q_p(o_j))$ and the perspective argument of Lemma 1(i) does not apply; concavity of the grid values is then not guaranteed (it may still hold numerically). Replacing the rule by a Bayes-consistent one restores (iii): for instance fixed nodes o_j with weights ω_j of a quadrature for Lebesgue measure on a truncated range (trapezoid, or Gauss–Hermite nodes o_j with $\omega_j = w_j/\phi(o_j)$), applied to the exact predictive density $\Lambda_p(o_j)$. (b) On a triangulated grid of $\mathcal{S}_2$, the piecewise-linear interpolant of the restriction of a concave function need not be concave. Take two adjacent cells ABC and ACD of the triangulation forming a parallelogram $ABCD$ (so $D = A + C - B$ and the diagonals AC, BD bisect each other at O). Let ℓ be the affine function with $\ell(A) = 1$, $\ell(C) = -1$, $\ell(B) = 0$; then $\ell(D) = \ell(A) + \ell(C) - \ell(B) = 0$ and $\ell(O) = 0$. The function $g = -\ell^2$ is concave, with $g(A) = g(C) = -1$ and $g(B) = g(D) = 0$. Its interpolant is affine on each cell, equal to -1 along the shared edge AC , hence -1 at O ; along the segment BD it takes the values $0, -1, 0$ at B, O, D and is linear on $[B, O]$ and $[O, D]$, which is not concave. So no concavity of the interpolated three-state grid values is claimed. $\square$

Proof of Theorem 1. Write $V_t(\cdot, v; \theta)$ and $Q_t(\cdot, v, a; \theta)$ for the value and Q -functions of Definition 7 at a parameter value θ , all other parameters fixed.

As in the proof of Proposition 1, every expectation in Definition 7 is a Bayes operator T_λ (Lemma 1) for the experiment of the action: λ^{monitor} , $\lambda^{\text{throttle}}$ and λ^v (the verification outcome and, if $\omega = 1$, ℓ_v monitoring statistics).

Step 1: the experiments are garblings. (a) *In κ .* Let $0 \leq \kappa < \kappa' \leq 1$ and $r = \kappa/\kappa' \in [0, 1]$. In the two-state model, $\lambda_M^{\text{throttle}}$ at κ is $\mathcal{N}(\pm \frac{1}{2}\kappa d, 1)$, which is the image of $\mathcal{N}(\pm \frac{1}{2}\kappa' d, 1)$ under the kernel of Lemma 1(iv) with this r , the same kernel for K and F . In the three-state model, $\mathcal{N}(\kappa\mu_M, I_k)$ is the image of $\mathcal{N}(\kappa'\mu_M, I_k)$ under the same kernel in $\mathbb{R}^k$, for every M . The experiments λ^{monitor} and λ^v do not depend on κ . (b) *In d (two-state model).* Let $0 \leq d < d'$ and $r = d/d' \in [0, 1]$. Then $\mathcal{N}(\pm \frac{1}{2}d, 1)$ is the image of $\mathcal{N}(\pm \frac{1}{2}d', 1)$ under the kernel of Lemma 1(iv), and $\mathcal{N}(\pm \frac{1}{2}\kappa d, 1)$ is the image of $\mathcal{N}(\pm \frac{1}{2}\kappa d', 1)$ under the same kernel (if $\kappa = 0$ both are $\mathcal{N}(0, 1)$ and the identity kernel serves). For **verify** with $\omega = 1$, apply the kernel to each of the ℓ_v statistics independently and leave z unchanged; the product of Markov kernels is a Markov kernel that does not depend on M , and it maps $\lambda^v(d')$ to $\lambda^v(d)$ because the densities are products (Assumptions 2–3). With $\omega = 0$, λ^v does not depend on d .

Step 2: induction. Let $\theta < \theta'$ stand for $\kappa < \kappa'$ (two-state or three-state model) or for $d < d'$ (two-state model). We show $V_t(\cdot, v; \theta) \geq V_t(\cdot, v; \theta')$ for all t, v by backward induction. At $t = T$ both equal the terminal risk. Let $t < T$ and assume the inequality for all $s > t$ and both v . The stage costs, the terminal costs and the available action set do not depend on θ . For an action with experiment $\lambda(\theta)$ and continuation $V_s(\cdot, v')$,

$$T_{\lambda(\theta)}V_s(\cdot, v'; \theta) \geq T_{\lambda(\theta)}V_s(\cdot, v'; \theta') \geq T_{\lambda(\theta')}V_s(\cdot, v'; \theta').$$

The first inequality holds because T_λ is monotone (its integrand is $\Lambda_p \geq 0$ times f) and by the induction hypothesis. The second is Lemma 1(iii), applicable because $\lambda(\theta)$ is a garbling of $\lambda(\theta')$ (Step 1; when the experiment does not depend on θ it is an equality) and $V_s(\cdot, v'; \theta')$ is concave and continuous (Proposition 1). Hence $Q_t(\cdot, v, a; \theta) \geq Q_t(\cdot, v, a; \theta')$ for every available a , and taking minima, $V_t(\cdot, v; \theta) \geq V_t(\cdot, v; \theta')$. This proves (i) and (ii).

(iii) The optimal value without **throttle**, V_0^{KSV} (Definition 9(iii): every action but **throttle**), is computed by the recursion of Definition 7 with **throttle** removed, which involves no quantity depending on κ . Every policy of that restricted recursion is a policy of the full one, so $V_0(p_0, 0; \kappa) \leq V_0^{\text{KSV}}(p_0, 0)$ and $G(\kappa) \geq 0$. By (i), $\kappa \mapsto V_0(p_0, 0; \kappa)$ is non-increasing, so G is non-decreasing. Let $U = \{\kappa \in [0, 1] : G(\kappa) > 0\}$. If $\kappa \in U$ and $\kappa < \kappa'' \leq 1$, then

$G(\kappa'') \geq G(\kappa) > 0$, so U is an up-set of $[0, 1]$, i.e. empty, $(\kappa^*, 1]$ or $[\kappa^*, 1]$ with $\kappa^* = \inf U$ (Definition 10, $\kappa^* = 1$ when U is empty).

(iv) If $\kappa < \kappa^*$, then $\kappa \notin U$ and $G(\kappa) = 0$: the optimal value of the restricted recursion equals $V_0(p_0, 0; \kappa)$, so an optimal policy of the restricted recursion, which never throttles, is optimal from p_0 in the full model. If $\kappa > \kappa^*$, there is $\kappa'' \in U$ with $\kappa'' < \kappa$ (definition of the infimum), so $G(\kappa) \geq G(\kappa'') > 0$. Let π be a policy that throttles with probability zero from p_0 . Define π' by replacing **throttle** with **monitor** at every decision state where π throttles. Up to the first time π would throttle, π and π' take the same actions and induce the same law of histories, and that first time is reached with probability zero; so the two policies generate the same law of the cost with probability one, and $J^\pi(p_0) = J^{\pi'}(p_0) \geq V_0^{\text{KSV}}(p_0, 0) > V_0(p_0, 0; \kappa)$. Hence π is not optimal from p_0 , and every policy optimal from p_0 throttles with positive probability. The three-state versions of (iii) and (iv) follow in the same way from the three-state form of (i).

Remark (what (iv) does not say). (iv) does not say that for $\kappa < \kappa^*$ the tie-broken optimal policy of Definition 7 never throttles: at a belief where **throttle** ties with the optimal action and ranks lower in the tie-breaking order (it ranks below **kill**), the tie-broken policy throttles. Such ties do not change the expected cost. $\square$

Proof of Proposition 2. Write $Q^k(p) = pC_K + (1-p)C_F$ and $Q^l(p) = pH_K + (1-p)H_F$ for the terminal Q -values (they do not depend on t or v). Then

$$Q^l(p) - Q^k(p) = p(H_K - C_K) - (1-p)(C_F - H_F) = pa - (1-p)b = (a+b)(p - q^*),$$

with $a, b > 0$ by Assumption 1, so $q^* = b/(a+b) \in (0, 1)$, $Q^l < Q^k$ on $[0, q^*)$, and $Q^l > Q^k$ on $(q^*, 1]$.

(i) At $p = 1$, for any experiment λ of Definition 7, $\Lambda_1 = \lambda_K$ and the posterior is 1 wherever $\Lambda_1 > 0$; hence $T_\lambda f(1) = f(1) \int \lambda_K d\nu = f(1)$. By backward induction $V_t(1, v) = C_K$: at $t = T$, $V_T(1, v) = \min\{C_K, H_K\} = C_K$ (Assumption 1); for $t < T$, **kill** gives C_K , **leave** gives $H_K > C_K$, and **monitor**, **throttle**, **verify** give $r_K + C_K$, $\delta + \rho r_K + C_K$, $c_v + \ell_v r_K + C_K$, each at least C_K because the stage costs are non-negative (Assumption 1). At $p = 0$ the same argument with $\Lambda_0 = \lambda_F$, posterior 0, stage costs 0, δ , c_v (no harm under F) and $C_F > H_F$ gives $V_t(0, v) = H_F$.

(ii) Let $D_t(p, v) = \min\{Q_t(p, v, a) : a \in \mathcal{A}_t(v), a \neq \text{kill}\}$, a minimum of finitely many concave continuous functions (Proposition 1(i)), hence concave and continuous. Then $\mathcal{K}_t(v) = \{p : D_t(p, v) - Q^k(p) \geq 0\}$ is a superlevel set

of a concave continuous function: a closed interval. It contains 1 by (i), since $Q^k(1) = C_K = V_t(1, v)$. So $\mathcal{K}_t(v) = [p_t^K(v), 1]$. The same argument with **leave** in place of **kill**, and $V_t(0, v) = H_F = Q^l(0)$, gives $\mathcal{L}_t(v) = [0, p_t^L(v)]$. If $p \in \mathcal{L}_t(v)$, then $Q^l(p) = V_t(p, v) \leq Q^k(p)$, so $p \leq q^*$; if $p \in \mathcal{K}_t(v)$, then $p \geq q^*$. So $p_t^L(v) \leq q^* \leq p_t^K(v)$ and $\mathcal{K}_t(v) \cap \mathcal{L}_t(v) \subseteq \{q^*\}$.

(iii) **leave** is the lowest rung of the tie-breaking order of Definition 7, so the tie-broken policy takes it exactly where it is optimal, on $\mathcal{L}_t(v)$. **kill** is the highest rung, so the policy takes it exactly where it is the unique optimal action, i.e. on $\{p : Q^k(p) < D_t(p, v)\}$, a strict superlevel set of a concave continuous function: an interval open relative to $[0, 1]$, contained in $\mathcal{K}_t(v)$. If $r_K > 0$ and $\delta + pr_K > 0$, then at $p = 1$ every other available action is strictly worse than **kill** by the computation in (i): $H_K > C_K$, $r_K > 0$, $\delta + pr_K > 0$, $c_v + \ell_v r_K > 0$ (as $\ell_v \geq 1$); so 1 is in the interval. (At $t = T$ the interval is $(q^*, 1]$.)

(iv) In the three-state model, the vertex e_M is absorbing for every Bayes update (the posterior of a point mass is that point mass wherever the predictive density is positive), so the argument of (i) gives $V_t(e_K, v) = C_K$, attained by **kill**, and $V_t(e_F, v) = H_F$, $V_t(e_B, v) = H_B$, attained by **leave** (stage costs under F and B are 0, $\delta, c_v \geq 0$ since $r_F = r_B = 0$, and $C_F > H_F$, $C_B > H_B$). The set where **kill** is optimal is the superlevel set $\{D_t - Q^k \geq 0\}$ with D_t concave (Proposition 1(ii)) and Q^k affine on $\mathcal{S}_2$: convex, and it contains e_K . The set where **leave** is optimal is convex by the same argument and contains e_F and e_B , hence the edge $[e_F, e_B]$. $\square$

Proof of Proposition 3. Throughout, $R(p) = \min\{Q^k(p), Q^l(p)\}$ with Q^k, Q^l the terminal Q -values of Definition 7; this is Paper 2's passive risk (Definition 1 there). Paper 2's verify-first risk with $r_K \tau$ read as $\ell_v r_K$ is

$$R_v(p) = c_v + \ell_v pr_K + \sum_{z \in \{+, -\}} P(z) R(p^z),$$

with $P(z)R(p^z)$ read as $\min\{P(z, K)C_K + P(z, F)C_F, P(z, K)H_K + P(z, F)H_F\}$ (zero when $P(z) = 0$). With the verification outcome as the observation ($\lambda_M(z) = P(Z = z \mid M)$, counting measure), this reading is exactly Lemma 1's $T_\lambda R(p)$: the integrand $\Lambda_p(z) R(q_p(z))$ is $F(P(z, K), P(z, F))$ for the perspective F of R , and $F(u, w) = \min\{uC_K + wC_F, uH_K + wH_F\}$ because R is a minimum of affine functions. So $R_v(p) = c_v + \ell_v pr_K + T_\lambda R(p)$.

(i) Under P2 (Definition 9(iv)), if $p \notin S$ the operator takes the better terminal action at p , at cost $R(p)$. If $p \in S$ it verifies (available because

$T \geq \ell_v$): the ℓ_v periods in the **verifying** posture cost r_M each, i.e. $\ell_v r_K$ under K and 0 under F (Definition 6), plus c_v ; order flow is not used, so the terminal action is the better one at p^Z , with expected terminal cost $\sum_z P(z)R(p^z)$. The expected cost is $R_v(p)$. Since $S = \{R_v < R\}$, $J^{P2}(p) = \min\{R(p), R_v(p)\}$. This does not depend on d .

(ii) Let $d = 0$. Then $d_{\text{monitor}} = d_{\text{throttle}} = 0$: the order-flow likelihoods under K and F coincide, every order-flow update leaves the belief unchanged (Definition 5), and $T_{\lambda^x} f(p) = f(p)$ for both postures. With $\omega = 1$ the ℓ_v monitoring statistics during verification are likewise uninformative, so $p'' = p^Z$ whatever ω . $v = 1$. By backward induction $V_t(p, 1) = R(p)$: at $t = T$ by definition; for $t < T$, **kill** and **leave** give $Q^k(p)$, $Q^l(p)$; **monitor** gives $pr_K + V_{t+1}(p, 1) = pr_K + R(p) \geq R(p)$; **throttle** gives $\delta + \rho pr_K + R(p) \geq R(p)$ (Assumption 1: $r_K, \delta \geq 0$, $\rho \in [0, 1]$); **verify** is not available. The minimum is $R(p)$. $v = 0$. If $t + \ell_v > T$, **verify** is not available at t or at any later time, and the same induction gives $V_t(p, 0) = R(p)$. If $t + \ell_v \leq T$, assume the claim for $t + 1$. **verify** gives $c_v + \ell_v pr_K + \mathbb{E}[V_{t+\ell_v}(p^Z, 1)] = c_v + \ell_v pr_K + T_{\lambda} R(p) = R_v(p)$. **monitor** gives $pr_K + V_{t+1}(p, 0) \geq V_{t+1}(p, 0) \geq \min\{R(p), R_v(p)\}$, because $V_{t+1}(p, 0)$ is $\min\{R, R_v\}(p)$ or $R(p)$; **throttle** likewise. The terminal actions give at least $R(p)$. So $V_t(p, 0) = \min\{R(p), R_v(p)\}$. At $t = 0$, $0 + \ell_v \leq T$, so $V_0(p, 0) = \min\{R(p), R_v(p)\} = J^{P2}(p)$ by (i).

(iii) Let $d = 0$, $r_K > 0$, $\delta + \rho r_K > 0$ (equivalently $\delta > 0$ or $\rho > 0$). For $p \in (0, 1]$, the bounds of (ii) are strict: **monitor** costs at least $pr_K + \min\{R, R_v\}(p) > V_0(p, 0)$, and **throttle** at least $\delta + \rho pr_K + \min\{R, R_v\}(p) > V_0(p, 0)$. Hence at $t = 0$ the optimal actions are among **leave**, **verify**, **kill**. (At $p = 0$, **leave** is optimal, $V_0(0, 0) = H_F$, and it is the lowest rung, so it is taken; $0 \notin S$ because $R_v(0) = c_v + R(0) \geq R(0)$.) If $p \in S$, **verify** is the unique optimal action. If $p \notin S$ and $p \leq q^*$, **leave** is optimal ($R(p) = Q^l(p)$, Proposition 2's identity) and it is the lowest rung. If $p \notin S$ and $p > q^*$, **kill** is optimal and **leave** is not; **verify** is optimal too exactly when $R_v(p) = R(p)$, and then Definition 7's tie-breaking chooses **verify** (it ranks below **kill**), whereas Paper 2 resolves ties in favour of the passive action (its Definition 3). This is the stated exception; both choices have the same expected cost. After **verify**, at $t = \ell_v$ with $v = 1$ and belief p^Z , $V_{\ell_v}(p^Z, 1) = R(p^Z)$ by (ii). If $\ell_v = T$ only terminal actions are available. If $\ell_v < T$, **monitor** and **throttle** cost $p^Z r_K + R(p^Z)$ and $\delta + \rho p^Z r_K + R(p^Z)$, which exceed $R(p^Z)$ when $p^Z > 0$; when $p^Z = 0$, **leave** is optimal and is the lowest rung. So the policy takes the better terminal action at p^Z (**leave** on a tie, the lowest rung), as P2 does. Hence the tie-broken optimal policy is P2, up to the stated tie. If instead

$r_K = 0$ (or $\delta = \rho = 0$), **monitor** (or **throttle**) may attain $V_0(p, 0)$ and, ranking below **verify** and **kill** (resp. below **kill**), be chosen; the value identity (ii) is unaffected.

(iv) By Theorem 1(ii), $V_0(p, 0; d) \leq V_0(p, 0; 0)$ for every $d \geq 0$, and $V_0(p, 0; 0) = J^{P^2}(p)$ by (ii); J^{P^2} does not depend on d by (i).

The horizon condition. **verify** at t needs $t + \ell_v \leq T$ (Definition 7). If $T < \ell_v$, it is never available, and the argument for $v = 1$ gives $V_0(p, 0) = R(p)$ when $d = 0$. At $T = 1$ the embedding therefore holds only with $\ell_v = 1$; with $\ell_v \geq 2$ it needs $T \geq \ell_v$. $\square$

Proof of Proposition 6. Throughout, $\mathcal{A}_t(v)$ is the action set at (t, v) (Definition 7). It is *non-increasing in t* : at $t = T$ only **kill** and **leave** are available, and **verify** is available at $t < T$ iff $v = 0$ and $t + \ell_v \leq T$, a condition that holds at t whenever it holds at $t + 1$. The same is true of the restricted action sets of Definition 9.

(i) *Monotonicity in the remaining horizon.* We show $V_s \leq V_{s+1}$ for $s = T - 1, T - 2, \dots, 0$ by backward induction on s (for all p, v). For $s = T - 1$: **kill** and **leave** are available at $T - 1$, so $V_{T-1}(p, v) \leq \min\{Q(\text{kill}), Q(\text{leave})\} = R(p) = V_T(p, v)$. Let $s < T - 1$ and suppose $V_u \leq V_{u+1}$ for every u with $s < u < T$. For each $a \in \mathcal{A}_{s+1}(v) \subseteq \mathcal{A}_s(v)$ we compare $Q_s(p, v, a)$ with $Q_{s+1}(p, v, a)$:

- **kill, leave:** equal (they do not depend on the time).
- **monitor, throttle:** $Q_s = c + \mathbb{E}_y[V_{s+1}(p', v)] \leq c + \mathbb{E}_y[V_{s+2}(p', v)] = Q_{s+1}$. The stage cost c , the law of y and the update p' do not depend on the time, and $V_{s+1} \leq V_{s+2}$ by hypothesis.
- **verify** (if in $\mathcal{A}_{s+1}(v)$): $Q_s = c_v + \ell_v p r_K + \mathbb{E}[V_{s+\ell_v}(p'', 1)] \leq c_v + \ell_v p r_K + \mathbb{E}[V_{s+1+\ell_v}(p'', 1)] = Q_{s+1}$, by the hypothesis at $u = s + \ell_v > s$. If $s + \ell_v = T$ the hypothesis is not needed, since then **verify** is not in $\mathcal{A}_{s+1}(v)$.

Hence $V_s(p, v) = \min_{a \in \mathcal{A}_s(v)} Q_s \leq \min_{a \in \mathcal{A}_{s+1}(v)} Q_s \leq \min_{a \in \mathcal{A}_{s+1}(v)} Q_{s+1} = V_{s+1}(p, v)$. The argument uses only that the action sets shrink with time and that the stage costs and kernels do not depend on it, so it applies verbatim to $V^{\text{KS}}, V^{\text{KSV}}, V^{\text{KST}}$.

(ii) $\kappa = 0$. At $\kappa = 0$ the throttled statistic has the law $\mathcal{N}(0, 1)$ under both causes (Definition 3), so the update of Definition 5 gives $p' = p$ ($d_x y = 0$).

Hence

$$Q_t(p, v, \text{throttle}) = \delta + \rho pr_K + V_{t+1}(p, v) \geq \delta + \rho pr_K + V_t(p, v)$$

by (i). This is the first claim. For the second we show $V_t(\cdot, v; 0) = V_t^{\text{KSV}}(\cdot, v)$ by backward induction. Equality holds at $t = T$ (both are R). If it holds at every $u > t$, every $Q_t(\cdot, v, a)$ with $a \neq \text{throttle}$ coincides with its counterpart in the KSV^* recursion (the same expression with $V_u = V_u^{\text{KSV}}$ for $u > t$), so their minimum is V_t^{KSV} . Moreover

$$Q_t(p, v, \text{throttle}) = \delta + \rho pr_K + V_{t+1}^{\text{KSV}}(p, v) \geq V_{t+1}^{\text{KSV}}(p, v) \geq V_t^{\text{KSV}}(p, v)$$

by (i) for the restricted recursion ($\delta, \rho, r_K \geq 0$, Assumption 1). So $V_t = \min\{V_t^{\text{KSV}}, Q_t(\text{throttle})\} = V_t^{\text{KSV}}$. At $t = 0$ this is $G(0) = V_0^{\text{KSV}} - V_0 = 0$ (Definition 10).

(iii) $\delta > 0$. For $\kappa \in [0, 1]$, $t < T$ and v , put

$$\omega_t(\kappa) = \sup_{p \in [0, 1]} \left| \mathbb{E}_y \left[V_{t+1}^{\text{KSV}}(p'_\kappa, v) \right] - V_{t+1}^{\text{KSV}}(p, v) \right|,$$

where p'_κ is the update after a throttled statistic at factor κ and y has the predictive law at p . V^{KSV} does not depend on κ .

We claim $\omega_t(\kappa) \rightarrow 0$ as $\kappa \rightarrow 0$:

- $V_{t+1}^{\text{KSV}}(\cdot, v)$ is continuous on the compact $[0, 1]$ (step *Continuity of V^{KSV}* at the end of the proof), hence uniformly continuous with a modulus m , bounded because V_{t+1}^{KSV} is bounded.
- The logistic function is $\frac{1}{4}$ -Lipschitz, so $|p'_\kappa - p| \leq \frac{1}{4}\kappa d|y|$.
- Under the predictive law, $|y| \leq \frac{1}{2}\kappa d + |\xi|$ with $\xi \sim \mathcal{N}(0, 1)$, whatever p .
- Hence $\omega_t(\kappa) \leq \mathbb{E} \left[m \left(\frac{1}{4}\kappa d \left(\frac{1}{2}\kappa d + |\xi| \right) \right) \right]$, which tends to 0 by dominated convergence, uniformly in p .

Since T is finite and $\delta > 0$, there is $\kappa_0 \in (0, 1]$ with $\max_{t < T, v} \omega_t(\kappa) < \delta$ for all $\kappa \in (0, \kappa_0]$. It depends on d , T and the parameters of KSV^* , not on the prior.

Fix such a κ (the case $\kappa = 0$ is (ii)). We show by backward induction that $V_t(\cdot, v; \kappa) = V_t^{\text{KSV}}(\cdot, v)$ and that **throttle** is not optimal at any (t, p, v) .

Equality holds at T . If it holds at every $u > t$, the non-throttle Q -values coincide with those of KSV^* as in (ii), and

$$Q_t(p, v, \text{throttle}) = \delta + \rho pr_K + \mathbb{E}_y[V_{t+1}^{\text{KSV}}(p'_\kappa, v)] \geq \delta - \omega_t(\kappa) + V_{t+1}^{\text{KSV}}(p, v) > V_{t+1}^{\text{KSV}}(p, v) \geq V_t^{\text{KSV}}(p, v),$$

using $\rho pr_K \geq 0$ and (i). So **throttle** is strictly worse than the best other action, and $V_t = V_t^{\text{KSV}}$.

At $t = 0$, $G(\kappa) = 0$ for every $\kappa \in [0, \kappa_0]$, so $\kappa^* = \inf\{\kappa : G(\kappa) > 0\} \geq \kappa_0 > 0$ (Definition 10; G is non-decreasing by Theorem 1(iii), but this is not needed here).

Three-state model. Replace p by $\pi \in \mathcal{S}_2$, pr_K by $\pi(K)r_K$, and the logistic Lipschitz bound by the bound for the three-state posterior. For $\kappa \leq 1$ the posterior $\pi'(M) \propto \pi(M)\phi_k(y - \kappa\mu_M)$ has every likelihood ratio $\phi_k(y - \kappa\mu_M)/\phi_k(y - \kappa\mu_{M'})$ in $[e^{-c}, e^c]$ with $c = 2\kappa L\|y\| + \kappa^2 L^2$, $L = \max_M \|\mu_M\|$, hence $\pi'(M)/\pi(M) \in [e^{-c}, e^c]$ and $\|\pi' - \pi\|_1 \leq e^c - 1$, which again tends to 0 in expectation, uniformly in π . Continuity of V^{KSV} on $\mathcal{S}_2$ is the step below, three-state case. Steps (i)–(iii) are otherwise unchanged.

Continuity of V^{KSV} (closes the dependency gap of 2026-09-30). We show by backward induction that $V_t^{\text{KSV}}(\cdot, v)$ is continuous on $[0, 1]$ (on $\mathcal{S}_2$ in the three-state model) for every $t \leq T$ and v . $V_T^{\text{KSV}} = R$ is a minimum of two affine functions. Let $t < T$ and suppose $V_u^{\text{KSV}}(\cdot, v)$ continuous for all $u > t$ and v . The Q -values of the restricted recursion at t are:

- **kill, leave:** affine in p .
- **monitor:** $pr_K + (T_\lambda V_{t+1}^{\text{KSV}}(\cdot, v))(p)$, with λ_K, λ_F the densities of $\mathcal{N}(\pm \frac{1}{2}d, 1)$ with respect to Lebesgue measure; continuous by Lemma 1(ii).
- **verify** ($v = 0$, $t + \ell_v \leq T$): $c_v + \ell_v pr_K + (T_\lambda V_{t+\ell_v}^{\text{KSV}}(\cdot, 1))(p)$, where the observation is $o = (z, y_1, \dots, y_{\ell_v})$ if $\omega = 1$ and $o = z$ if $\omega = 0$, on $O = \{+, -\} \times \mathbb{R}^{\ell_v}$ (resp. $\{+, -\}$) with counting times Lebesgue measure, and $\lambda_M(o) = P(Z = z \mid M) \prod_i \phi(y_i \mp \frac{1}{2}d)$ (the product is absent if $\omega = 0$). These are probability densities, and by Assumption 3 and Assumption 2 they are the joint law of the verification outcome and the order flow of the **verifying** periods given M ; p'' is the Bayes posterior after o (Definition 5, the updates in sequence give the joint Bayes posterior). Continuous by Lemma 1(ii).

$V_t^{\text{KSV}}(\cdot, v)$ is the minimum of finitely many continuous functions over an action set that does not depend on p , hence continuous. The three-state

case is the same with the three-state form of Lemma 1(ii) (last sentence of Lemma 1) and $\lambda_M(o) = P(Z = z \mid M) \prod_i \phi_k(y_i - \mu_M)$. This uses only Lemma 1(ii) and Definitions 3–7, 9; Proposition 1 is not needed. It agrees with the closure proposed on 2026-09-30, which remains valid as a second route: by (ii), whose proof uses (i) and no continuity, $V_t^{\text{KSV}}(\cdot, v) = V_t(\cdot, v; 0)$, the value of the recursion of Definition 7 at $\kappa = 0$, and Proposition 1 as stated covers every $\kappa \geq 0$ (Definition 3 allows $\kappa = 0$; Assumption 4 is not among its hypotheses). Either route closes the gap; the statement's reference to Proposition 1 is correct through the second. $\square$

Proof of Proposition 4. Step 0: mean plus common noise. Fix a schedule Θ and a cause M . Under D2 and D9 (with either throttle), the state and observation recursions are linear in (x_0, w, ε) and in the deterministic injection terms, with the same matrices A and C under both causes and all schedules. Let x_t^0 solve $x_{t+1}^0 = Ax_t^0 + w_t$ from x_0 . Then $y_t = \mu_t(M, \Theta) + Cx_t^0 + \varepsilon_t$, where $\mu_t(M, \Theta)$ is deterministic and collects the injection terms. By (A1)–(A3), the noise $N_W = (Cx_t^0 + \varepsilon_t)_{t \in W}$ is a Gaussian vector whose law does not depend on M or Θ ; it has mean zero when $x_0 = 0$ (A2), and for a Gaussian x_0 whose law does not depend on the instance its mean is common to both causes and cancels in every difference below (absorb it into μ). Its covariance Σ_W is at least the block-diagonal matrix with blocks $\Sigma \succ 0$ (A6), since ε is independent of x^0 ; so $\Sigma_W \succ 0$. Hence $Y_W \sim \mathcal{N}(\mu_W(M, \Theta), \Sigma_W)$, the law of Y_W is the same under K and F if and only if $e_W(\Theta) := \mu_W(K, \Theta) - \mu_W(F, \Theta) = 0$, and $d_W(\Theta) = (e_W(\Theta)^\top \Sigma_W^{-1} e_W(\Theta))^{1/2}$ (Paper 1, Definition D5).

Mean recursions. Let m_t be the mean of the state variable x_t of D2/D9 minus that of x_t^0 , and $\iota_t(M) = \theta_M b_M$ for $t \geq \tau$, 0 for $t < \tau$. Then $m_t = 0$ for $t \leq \tau$, and:

- D2, any schedule: $m_{t+1} = Am_t$, so $m_t = 0$ for all t , and $\mu_t = C s_t \iota_t(M)$ with $s_t = r$ if $t \in \Theta$ (scaling throttle) and $s_t = 1$ otherwise;
- D9, scaling throttle: $m_{t+1} = A(m_t + s_t \iota_t(M))$, $\mu_t = C(m_t + s_t \iota_t(M))$;
- D9, feedback-suppressing throttle: $m_{t+1} = A(m_t + \iota_t(M))$ for $t \notin \Theta$, $m_{t+1} = Am_t$ for $t \in \Theta$; $\mu_t = C(m_t + \iota_t(M))$ in both cases.

All are linear in $\iota(M)$, so the mean differences e_t satisfy the same recursions with $\iota_t(M)$ replaced by $\Delta_0 \mathbb{K}[t \geq \tau]$.

(i) If $\Theta \supseteq \{\tau, \dots, t_W\}$, then $s_t = r$ for every t with $\tau \leq t \leq t_W$, and $\iota_t = 0$ for $t < \tau$. By linearity, for $t \leq t_W$ the difference recursion under Θ is the one under monitoring with Δ_0 replaced by $r\Delta_0$, so $e_W(\Theta) = r e_W(\emptyset)$ and $d_W(\Theta) = r d_W(\emptyset)$. Under D2, $e_t = C s_t \Delta_0$ for $t \geq \tau$ and 0 before, so $\Theta \supseteq W \cap \{t \geq \tau\}$ suffices.

(ii) D9, feedback-suppressing throttle, $C\Delta_0 = 0$, $\Theta \supseteq \{\tau, \dots, t_W\}$. For $t \leq t_W$ the difference of states obeys $m_{t+1} = A m_t$ from $m_\tau = 0$, so $m_t = 0$ and $e_t = C(0 + \Delta_0) = 0$ for $\tau \leq t \leq t_W$, and $e_t = 0$ for $t < \tau$. Hence $e_W(\Theta) = 0$: the laws coincide and $d_W(\Theta) = 0$. Under monitoring (D9), $m_{\tau+k} = \sum_{j=1}^k A^j \Delta_0$ and $e_{\tau+k} = C \sum_{j=0}^k A^j \Delta_0 = C \sum_{j=1}^k A^j \Delta_0$ (using $C\Delta_0 = 0$), and $e_t = 0$ for $t \leq \tau$. So $e_W(\emptyset) \neq 0$, i.e. $d_W(\emptyset) > 0$, if and only if $C \sum_{j=1}^k A^j \Delta_0 \neq 0$ for some $k \geq 1$ with $\tau + k \in W$. If $W \cap \{t \geq \tau\} = \{\tau, \dots, \tau + H - 1\}$, the relevant partial sums are $E_k = \sum_{j=1}^k C A^j \Delta_0$, $1 \leq k \leq H - 1$ (the time τ itself has $e_\tau = C\Delta_0 = 0$). With $E_0 = 0$, they all vanish if and only if every increment $E_k - E_{k-1} = C A^k \Delta_0$, $1 \leq k \leq H - 1$, vanishes. So $d_W(\emptyset) > 0$ if and only if $C A^j \Delta_0 \neq 0$ for some $1 \leq j \leq H - 1$. When $d_W(\emptyset) > 0$, $\kappa_W(\Theta) = 0$.

(iii) Late start, $\Theta = \{t_0, t_0 + 1, \dots\}$, $t_0 > \tau$, model D9. Differences of states start from $m_\tau = 0$. *Scaling throttle.* Let $u_t = m_t(\Theta) - r m_t(\emptyset)$ (differences between causes throughout). From the recursions, $u_\tau = 0$ and $u_{t+1} = A(u_t + (s_t - r)\Delta_0)$ with $s_t = 1$ for $\tau \leq t < t_0$ and $s_t = r$ for $t \geq t_0$. Hence $u_{t_0} = (1 - r) \sum_{j=1}^{t_0-\tau} A^j \Delta_0 = (1 - r) D_{t_0}$ and $u_t = (1 - r) A^{t-t_0} D_{t_0}$ for $t \geq t_0$. The mean differences of y_t satisfy $e_t(\Theta) - r e_t(\emptyset) = C(u_t + (s_t - r)\Delta_0) = (1 - r) C A^{t-t_0} D_{t_0}$ for $t \geq t_0$. *Feedback-suppressing throttle.* For $\tau \leq t < t_0$ the differences follow D9, so $m_{t_0} = D_{t_0}$; for $t \geq t_0$, $m_{t+1} = A m_t$, so $m_t = A^{t-t_0} D_{t_0}$ and $e_t(\Theta) = C A^{t-t_0} D_{t_0} + C\Delta_0$. *Example.* $n = 2$, $m = 1$, $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\Delta_0 = (0, 1)^\top$, so $C\Delta_0 = 0$ and $C A \Delta_0 = 1$; $t_0 = \tau + 1$. Then $D_{t_0} = A \Delta_0 = (1, 0)^\top$ and $C D_{t_0} = 1$. Feedback-suppressing: $e_{t_0}(\Theta) = 1 \neq 0$, so every window $W \subseteq \Theta$ containing t_0 has $e_W(\Theta) \neq 0$ and $d_W(\Theta) > 0$. Scaling: under monitoring $e_{t_0}(\emptyset) = C(\Delta_0 + A \Delta_0) = 1$, and $e_{t_0}(\Theta) = r + (1 - r) = 1$, so for $W = \{t_0\}$, $\kappa_W(\Theta) = 1 \neq r$ when $r < 1$.

A further caution for (iii), not needed for the statement. Even when the marginal mean difference of the throttled observations vanishes, throttled observations correlated with earlier ones (through x^0 , when $Q \neq 0$) carry information about M jointly with them: for $Y = (Y_1, Y_2)$ with mean difference $(e_1, 0)$, $d^2 = e_1^\top (\Sigma^{-1})_{11} e_1 = e_1^\top (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} e_1 \geq e_1^\top \Sigma_{11}^{-1} e_1$. this

paper's κ is a per-period factor under conditionally independent statistics (Assumption 2); κ_W of a window is the matching quantity only when the window starts at the onset, as in (i) and (ii).

(iv) Let $n = m = 1$, $C = 1$, $Q = 0$, $x_0 = 0$, $\Sigma = \sigma^2 > 0$, $\Delta_0 \neq 0$ (so $C\Delta_0 \neq 0$), $W = \{\tau, \tau + 1\}$, and a feedback-suppressing throttle with $\Theta \supseteq W$. Then $x^0 \equiv 0$, $\Sigma_W = \sigma^2 I_2$. Under monitoring (D9) $e_\tau = \Delta_0$ and $e_{\tau+1} = (1 + A)\Delta_0$; under the throttle (D2 dynamics) $e_\tau = e_{\tau+1} = \Delta_0$. Hence

$$\kappa_W^2 = \frac{2 \Delta_0^2 / \sigma^2}{(1 + (1 + A)^2) \Delta_0^2 / \sigma^2} = \frac{2}{1 + (1 + A)^2},$$

which is below one for $A > 0$, equal to one for $A = 0$, and above one for $-2 < A < 0$. Explicit instances: $A = 1$ gives $\kappa_W = \sqrt{2/5} < 1$; $A = 0$ gives $\kappa_W = 1$; $A = -1$ gives $\kappa_W = \sqrt{2} > 1$ (under monitoring $e_{\tau+1} = 0$, so the second observation carries no mean difference, while the throttle keeps it at Δ_0). So the model does not sign $1 - \kappa_W$ for pairs with $C\Delta_0 \neq 0$, and $\kappa \leq 1$ (Assumption 4) is a restriction of this paper's model. $\square$

References

- Bashendy, M., Tantawy, A., Erradi, A., 2023. Intrusion response systems for cyber-physical systems: A comprehensive survey. *Computers & Security* 124, 102984. doi:10.1016/j.cose.2022.102984.
- Blackwell, D., 1953. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics* 24, 265–272. doi:10.1214/aoms/1177729032.
- Bongaerts, D., De Luca, S.D., Van Achter, M., 2024. Circuit breakers and market runs. *Review of Finance* 28, 1953–1989. doi:10.1093/rof/rfae029.
- Campbell, S., Gaitsgori, G., Groenewald, R., 2025. A sequential testing problem with signal control. arXiv preprint 2509.18209. URL: <https://arxiv.org/abs/2509.18209>, arXiv:2509.18209.
- Cao, F., Jia, F., He, X., 2024. Input design for active fault detection: Reconciling system control objectives. *IEEE Transactions on Cybernetics* 54, 3931–3942. doi:10.1109/TCYB.2023.3331971.

- Chen, H., Petukhov, A., Wang, J., Xing, H., 2024. The dark side of circuit breakers. *The Journal of Finance* 79, 1405–1455. doi:10.1111/jofi.13310.
- Chernoff, H., 1959. Sequential design of experiments. *The Annals of Mathematical Statistics* 30, 755–770. doi:10.1214/aoms/1177706205.
- Guo, Y., He, X., 2026. Integrated design for active fault diagnosis and fault-tolerant optimal control for stochastic systems with non-convex input constraints. *Automatica* 188, 112960. doi:10.1016/j.automatica.2026.112960.
- Hammar, K., Dhir, N., Stadler, R., 2026. Optimal defender strategies for CAGE-2 using causal modeling and tree search. *IEEE Transactions on Network and Service Management* doi:10.1109/TNSM.2026.3735865. early access.
- Hammar, K., Li, T., Stadler, R., Zhu, Q., 2025. Adaptive security response strategies through conjectural online learning. *IEEE Transactions on Information Forensics and Security* 20, 4055–4070. doi:10.1109/TIFS.2025.3558600.
- Hammar, K., Stadler, R., 2022. Intrusion prevention through optimal stopping. *IEEE Transactions on Network and Service Management* 19, 2333–2348. doi:10.1109/TNSM.2022.3176781.
- Hammar, K., Stadler, R., 2024. Learning near-optimal intrusion responses against dynamic attackers. *IEEE Transactions on Network and Service Management* 21, 1158–1177. doi:10.1109/TNSM.2023.3293413.
- Hammar, K., Stadler, R., 2026. Online identification of IT systems through active causal learning. *IEEE Transactions on Network and Service Management* doi:10.1109/TNSM.2026.3736090. early access.
- Harrison, J.M., Sunar, N., 2015. Investment timing with incomplete information and multiple means of learning. *Operations Research* 63, 442–457. doi:10.1287/opre.2015.1344.
- Hou, Y., Bidkhori, H., Banerjee, T., 2025. Robust quickest change detection with sampling control. *IEEE Transactions on Signal Processing* 73, 3588–3597. doi:10.1109/TSP.2025.3609067.

- Jafar, M.T., Yang, L.X., Li, G., Zhu, Q., Gan, C., Yang, X., 2024. Malware containment with immediate response in IoT networks: An optimal control approach. *Computer Communications* 228, 107951. doi:10.1016/j.comcom.2024.107951.
- Kartik, D., Nayyar, A., Mitra, U., 2022. Fixed-horizon active hypothesis testing. *IEEE Transactions on Automatic Control* 67, 1882–1897. doi:10.1109/TAC.2021.3090742.
- Kazeminajafabadi, A., Imani, M., 2023. Optimal monitoring and attack detection of networks modeled by Bayesian attack graphs. *Cybersecurity* 6, 22. doi:10.1186/s42400-023-00155-y.
- Kazeminajafabadi, A., Imani, M., 2024. Optimal joint defense and monitoring for networks security under uncertainty: A POMDP-based approach. *IET Information Security* 2024, 7966713. doi:10.1049/2024/7966713.
- Krishnamurthy, V., 2025. *Partially Observed Markov Decision Processes*. 2 ed., Cambridge University Press. doi:10.1017/9781009449441.
- Lera, S.C., Sornette, D., Ulmann, F., 2024. Price dynamics with circuit breakers. *Quantitative Finance* 24, 1711–1724. doi:10.1080/14697688.2024.2429421.
- Liu, M., Zhao, C., Zhang, Z., Deng, R., 2022. Explicit analysis on effectiveness and hiddenness of moving target defense in AC power systems. *IEEE Transactions on Power Systems* 37, 4732–4746. doi:10.1109/TPWRS.2022.3152801.
- Lovejoy, W.S., 1987. Some monotonicity results for partially observed Markov decision processes. *Operations Research* 35, 736–743. doi:10.1287/opre.35.5.736.
- Lubenia, P.V.N., Banerjee, T., 2026. Quickest change detection with cost-constrained experiment design. *IEEE Transactions on Information Theory* 72, 7829–7847. doi:10.1109/TIT.2026.3723839.
- Mejrissi, C., 2026a. Attack–fault ambiguity against a strategic attacker: The stealth–harm frontier, exploitable commitments, and the verification margin. Working paper.

- Mejrissi, C., 2026b. Attack–fault confounding at the electronic trading gateway: Fragility under state feedback, the cost of ambiguity over the observation horizon, and a retrospective study of summary matching. Working paper.
- Mejrissi, C., 2026c. Verification as a costed action: When checking beats containing under attack–fault ambiguity. Working paper.
- Mesbah, A., 2018. Stochastic model predictive control with active uncertainty learning: A survey on dual control. *Annual Reviews in Control* 45, 107–117. doi:10.1016/j.arcontrol.2017.11.001.
- Miehling, E., Rasouli, M., Teneketzis, D., 2018. A POMDP approach to the dynamic defense of large-scale cyber networks. *IEEE Transactions on Information Forensics and Security* 13, 2490–2505. doi:10.1109/TIFS.2018.2819967.
- Mo, Y., Weerakkody, S., Sinopoli, B., 2015. Physical authentication of control systems: Designing watermarked control inputs to detect counterfeit sensor outputs. *IEEE Control Systems Magazine* 35, 93–109. doi:10.1109/MCS.2014.2364724.
- Moscarini, G., Smith, L., 2001. The optimal level of experimentation. *Econometrica* 69, 1629–1644. doi:10.1111/1468-0262.00259.
- Naghshvar, M., Javidi, T., 2013. Active sequential hypothesis testing. *The Annals of Statistics* 41, 2703–2738. doi:10.1214/13-AOS1144.
- Naha, A., Teixeira, A., Ahlén, A., Dey, S., 2023. Quickest physical watermarking-based detection of measurement replacement attacks in networked control systems. *European Journal of Control* 71, 100804. doi:10.1016/j.ejcon.2023.100804.
- Nguyen, T.T., Reddi, V.J., 2023. Deep reinforcement learning for cyber security. *IEEE Transactions on Neural Networks and Learning Systems* 34, 3779–3795. doi:10.1109/TNNLS.2021.3121870.
- Nitinawarat, S., Atia, G.K., Veeravalli, V.V., 2013. Controlled sensing for multihypothesis testing. *IEEE Transactions on Automatic Control* 58, 2451–2464. doi:10.1109/TAC.2013.2261188.

- Peskir, G., Shiryaev, A.N., 2006. Optimal Stopping and Free-Boundary Problems. Birkhäuser, Basel. doi:10.1007/978-3-7643-7390-0.
- Smallwood, R.D., Sondik, E.J., 1973. The optimal control of partially observable Markov processes over a finite horizon. Operations Research 21, 1071–1088. doi:10.1287/opre.21.5.1071.
- Straka, O., Punčochář, I., 2024. Active fault diagnosis for stochastic large scale systems under non-separable costs. Automatica 159, 111348. doi:10.1016/j.automatica.2023.111348.
- Tsopelakos, A., Fellouris, G., 2023. Sequential anomaly detection under sampling constraints. IEEE Transactions on Information Theory 69, 8126–8146. doi:10.1109/TIT.2022.3177142.
- Wang, X., Kim, M.H., Suardi, S., 2022. Herding and China’s market-wide circuit breaker. Journal of Banking & Finance 141, 106533. doi:10.1016/j.jbankfin.2022.106533.
- Zhang, K., Kasis, A., Parisini, T., Polycarpou, M.M., 2026. Threat discrimination between attacks and faults for a class of nonlinear systems. IEEE Transactions on Automatic Control 71, 946–961. doi:10.1109/TAC.2025.3604377.